# Conformalized Heteroscedastic Variant Effect Prediction with Local Coverage Guarantees

**Anonymous Author(s)**
Affiliation
Address
email

## Abstract

We introduce **HCCP** (Heteroscedastic Class-Conditional Conformal Prediction): a post-hoc framework pairing a learned variance head $\hat{\sigma}(x)$ with Mondrian-$(y \times \hat{\sigma}$-bin) calibration via a feature-dependent nonconformity score. For binary conformal classification under joint class imbalance and heteroscedasticity, existing variants trade class-conditional coverage against bin-local coverage, a Pareto wall sharpened by the finite-sample pointwise impossibility of Barber et al. [2020]. The gap binds in variant effect prediction (VEP), where TraitGym minority prevalence is $\pi_{\min} \approx 0.10$. Within the equi-bin Mondrian-$K$ family we prove a tight finite-sample $O(n^{-1/2})$ rate on the worst-cell coverage gap, matched *within this family* by an $\Omega(n^{-1/2})$ lower bound with $\pi_{\min}^{-1/2}$ explicit in the constant (Theorems 3, 4); the bound is dimension-free given $\hat{\sigma}$ is 1-D Lipschitz. On TraitGym [Benegas et al., 2025] at the recommended operating point ($K_{\text{eval}} = 3$ / $5$ for Mendelian / Complex; per-cell minority $\geq 100$), HCCP is the only single-fold partition we evaluated (across split CP, $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020], class-Mondrian, RLCP, weighted CP, and SC-CP at matched $(\hat{p}, \hat{\sigma})$) that holds $\text{cov}_{|Y=1} \geq 0.85$ and $\hat{\sigma}$-bin gap $\leq 0.20$ on both Mendelian ($n = 3{,}380$) and Complex ($n = 11{,}400$) under $B = 200$ chromosome-bootstrap CIs. A $K_{\text{eval}}$ sweep shows the Complex advantage is robust across $K_{\text{eval}} \in [2, 10]$, while on Mendelian weighted CP overtakes HCCP for $K_{\text{eval}} \geq 5$ (per-cell minority $< 70$); we replicate the analysis on ProteinGym [Notin et al., 2023] for cross-domain validation.

## 1 Introduction

Clinical variant interpretation drives consequential decisions [Richards et al., 2015], yet machine-learned variant effect predictors (VEPs) collapse evidence into a single probability [Kircher et al., 2014, Linder et al., 2025, Benegas et al., 2023, 2025] without explicit confidence. Conformal prediction guarantees distribution-free coverage [Vovk et al., 2005, 2003, Romano et al., 2019, Barber et al., 2021, 2023], yet Barber et al. [2020] rule out pointwise conditional coverage in finite samples. VEP therefore needs a method that (i) consumes a learned heteroscedastic head $\hat{\sigma}(x)$ as in DEGU [Zhou et al., 2026], (ii) stratifies calibration by both label and $\hat{\sigma}$-bin to recover as much conditioning as finite samples permit, and (iii) holds local coverage across distribution shifts. The gap binds for VEP: TraitGym minority prevalence sits at $\pi_{\min} \approx 0.10$, and chrom-LOO, trait-LOO, and cross-dataset shifts are intrinsic.

The technical question is: which Mondrian partition simultaneously delivers (i) class-conditional coverage under extreme imbalance and (ii) bin-local coverage under heteroscedasticity? Single-axis answers fail individually: $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020] attains local coverage in

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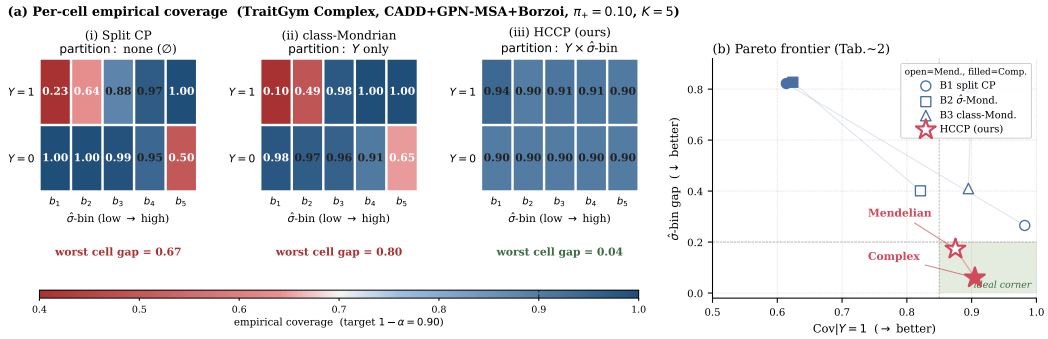


Figure 1: HCCP at a glance. **(a)** Per-cell empirical coverage on TraitGym Complex (CADD+GPN-MSA+Borzoi, $\pi_+$=0.10, $K$=5): three coverage grids over $Y \times \hat{\sigma}$-bin, colored by deviation from $1-\alpha$=0.90 (red = under, blue = over). **(i)** Split CP collapses on both ends of $\hat{\sigma}$ (worst cell 0.67). **(ii)** class-Mondrian [Sadinle et al., 2019] fixes the class budget but leaves $\hat{\sigma}$-bin variation within each class unconstrained (worst cell 0.80). **(iii)** HCCP partitions on $Y \times \hat{\sigma}$-bin and holds every cell within $\pm 0.04$ of target. The nonconformity score $s(x, y) = |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$ with per-cell calibration restores both (T2) class- and (T3) bin-conditional coverage, compatible with Barber et al. [2020]'s pointwise impossibility. **(b)** Pareto plot on (Cov$|Y$=1, $\hat{\sigma}$-bin gap) using Tab. 8 bootstrap means ($B$=200). HCCP is the only method in the ideal corner (cov $\geq 0.85$, gap $\leq 0.20$) on both datasets; B1, B2, B3 each fail at least one axis at $\pi_+$=0.10.

shape but loses minority-class coverage ($\text{cov}_{|Y=1} = 0.62$ on TraitGym Complex; the resulting minority-cell undercoverage drives its $\hat{\sigma}$-bin gap up to 0.83); class-Mondrian [Sadinle et al., 2019] attains class coverage but leaves a bin-local gap $13.4\times$ wider than the joint partition on Complex (bootstrap mean, $B = 200$). Within the equi-bin Mondrian-$K$ family, we prove the joint ($y \times \hat{\sigma}$-bin) Mondrian attains both with worst-cell coverage gap $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$ (Theorem 3) and a matching $\Omega(\sqrt{L_F R/(\pi_{\min} n)})$ finite-sample lower bound (Theorem 4); $\pi_{\min}$ enters *explicitly* as a $\pi_{\min}^{-1/2}$ constant.

**Contributions.**

1. **Tight finite-sample rate within the equi-bin Mondrian-$K$ class with $\pi_{\min}$ explicit in the constant** (Theorems 3, 4). To our knowledge this is the first finite-sample bound for class-conditional + heteroscedastic-local CP in which the minority-class probability $\pi_{\min}$ appears in the rate; the constant is dimension-free under a 1-D Lipschitz assumption on $\hat{\sigma}$. Concurrent imbalanced-CP work [Xie et al., 2025, Ding et al., 2023, Shi et al., 2024] provides marginal or class-cond rates without the $\pi_{\min}$-local product.

2. **HCCP, a post-hoc framework with an exchangeability-robust corollary.** HCCP transfers the heteroscedastic head of DEGU from MPRA regression to binary variant classification with finite-sample local coverage, pairing a nonconformity score $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ with Mondrian-($y \times \hat{\sigma}$-bin) calibration. T3 gives exact bin-conditional coverage at rate $1/(n_{k,b} + 1)$; T3$'$ (via Barber et al. [2023] Thm. 2 at the cell level) supplies a uniform finite-sample lower bound when our per-chromosome KS audit detects cell-level exchangeability violations on Mendelian.

3. **Empirical validation on TraitGym + ProteinGym + synthetic.** On TraitGym Mendelian / Complex at $\pi_{\min} = 0.10$ and the recommended operating point ($K_{\text{eval}} = 3$ / 5; per-cell minority $\geq 100$), HCCP is the only single-fold partition we evaluated (across split CP, $\hat{\sigma}$-Mondrian, class-Mondrian, RLCP, weighted CP, and SC-CP at matched $(\hat{p}, \hat{\sigma})$) that holds $\text{cov}_{|Y=1} \geq 0.85$ and $\hat{\sigma}$-bin gap $\leq 0.20$ on both datasets ($B = 200$ chrom-bootstrap CIs throughout). The conformal gain is decoupled from base-predictor accuracy (App. C.2). We replicate on ProteinGym [Notin et al., 2023] for cross-domain validation, and a synthetic $n$-sweep verifies the dimension-free $O(n^{-1/2})$ rate.

## 2 Related work

**Conformal prediction with conditioning.** Split / Mondrian CP [Vovk et al., 2005, 2003, Angelopoulos and Bates, 2022] provides marginal / group-conditional coverage; Barber et al. [2020] rule out pointwise conditional coverage. Normalized scores with a learned $\hat{\sigma}$ originate with Papadopoulos et al. [2008] and Lei et al. [2018] for regression; Boström and Johansson [2020] systematized Mondrian conformal regression with normalized scores; CQR [Romano et al., 2019] is the regression analog of our score. Recent local-CP work targets feature-space coverage at slower rates: **RLCP** [Hore and Barber, 2025] achieves $O(n^{-2/(d+2)})$ kernel-localized coverage (intractable at $d \approx 7{,}700$); **Self-Calibrating CP** [van der Laan, 2024] achieves $O(n^{-2/3})$ on the $\hat{\mu}$-axis via isotonic monotonicity [Deng et al., 2021], on a different axis from ours. Tibshirani et al. [2019], Gibbs and Candès [2021], Barber et al. [2023] cover distribution-shift robustness; we use Barber 2023 Thm. 2 for our T3′. De Wolf et al. [2025] analyze conditional validity of heteroscedastic CP for regression; we extend to classification with $\pi_{\min}$ explicit. Yao et al. [2026] (concurrent) prove a dimension-free $O(n^{-1/2})$ efficiency rate for split-CP quantile regression trained via SGD; their result establishes the rate on linear quantile reg. We contribute orthogonal axes — the partition-family analysis (equi-bin Mondrian-$K$), explicit $\pi_{\min}^{-1/2}$ constant, and within-class tight lower bound — not a faster rate. Plassier et al. [2025] target approximate conditional validity via density-ratio reweighting under TV-distance; an alternative to discrete $\hat{\sigma}$-bin Mondrian without finite-sample bin-conditional guarantees. Toccaceli [2026] (concurrent) identifies optimal bin count under covariate-sorted contiguous binning as an open problem; Theorem 3 resolves a special case for ($y \times \hat{\sigma}$-bin) Mondrian under class imbalance.

**Class-conditional and imbalanced CP.** Sadinle et al. [2019] introduced label-conditional CP (Mondrian-by-$y$); Shi et al. [2024], Ding et al. [2023] adapt class-conditional CP for many classes / small per-class samples; Xie et al. [2025] (concurrent) study open-set + imbalanced classification with reweighting and obtain a $O(n^{-1/2})$ marginal-coverage bound *without* $\pi_{\min}$-local product. Bairaktari et al. [2025] (concurrent) extend group-conditional CP to overlapping/fractional groups with minimax-optimal coverage subsuming class-conditional and Mondrian as static-group special cases; their framework requires pre-specified groups and does not cover our data-dependent ($y \times \hat{\sigma}$-bin) partition or provide a matching lower bound. None of these address the heteroscedastic axis: in our H2H ablation (Tab. 8), single-axis Mondrian predictors face a fundamental tradeoff in the imbalanced regime — class-Mondrian preserves $\mathrm{cov}_{|Y=1}$ but loses bin-local coverage (gap $0.42$–$0.80$, bootstrap mean), while $\hat{\sigma}$-Mondrian preserves local coverage but loses $\mathrm{cov}_{|Y=1}$ down to $0.62$ at $\pi_{\min} = 0.10$. HCCP's joint ($y \times \hat{\sigma}$-bin) Mondrian is, to our knowledge, the first construction simultaneously satisfying both objectives in this regime, with T5 quantifying the price as $O(\sqrt{L_F R/(\pi_{\min} n)})$.

**Heteroscedastic UQ and VEP benchmarks.** Nix and Weigend [1994] originated Gaussian NLL for learned variance; Lakshminarayanan et al. [2017] introduced deep ensembles. **DEGU** [Zhou et al., 2026] pioneered ensemble distillation to a Gaussian-uncertainty student for massively parallel reporter assay (MPRA) regression and motivated conformal prediction as future work; we extend its $\hat{\sigma}$ machinery to (a) variant classification and (b) finite-sample local coverage with adaptive partition granularity. We do not claim a head-to-head defeat of DEGU: DEGU is a regression pipeline on MPRA with a Keras CNN ensemble, whereas HCCP targets binary classification on TraitGym/ProteinGym; the "DEGU-lite" ablation in App. C reimplements the ensemble-distilled $\hat{\sigma}$ at our matched GBM backbone solely to isolate the choice of variance source, not to replicate DEGU's published pipeline. **TraitGym** [Benegas et al., 2025] standardizes VEP evaluation; features include CADD [Kircher et al., 2014], Borzoi [Linder et al., 2025], GPN-MSA [Benegas et al., 2023]; AlphaGenome [Avsec et al., 2026] is a recent foundation model used in our App. C ablation. CP applications in genomic medicine [Papangelou et al., 2025, Wang et al., 2025, Adegoke et al., 2025] target drug response or polygenic risk scores in a regression / clinical-screening setting; HCCP is, to our knowledge, the first CP framework with formal class-conditional and bin-local coverage guarantees for binary variant pathogenicity.

# 3 Problem formulation

## 3.1 Setup and assumptions

Let $\mathcal{D} = \{(X_i, Y_i, C_i)\}_{i=1}^n$ with $X_i \in \mathcal{X}$, $Y_i \in \{0,1\}$, and chromosome label $C_i \in \mathcal{C}$. We use the chrom-LOO protocol of Benegas et al. [2025]: a test chromosome $c^\star \in \mathcal{C}$ is held out and the calibration fold $\mathcal{D}^{(-c^\star)} := \{i : C_i \neq c^\star\}$ is used for both the aggregator $\hat{p}$ and the heteroscedastic head $\hat{\sigma}$. Test point $(X, Y) \sim P_{c^\star}$.

**Assumption 1** (A1 chrom-group exchangeability)**.** *Variants from the same chromosome are exchangeable:* $\{(X_i, Y_i) : C_i = c\} \overset{d}{=} \{(X_{\pi(i)}, Y_{\pi(i)}) : C_i = c\}$ *for any permutation* $\pi$ *within chromosome* $c$*.*

**Assumption 2** (A1′ chrom-wise i.i.d.)**.** $\{(X_i, Y_i) : C_i = c\} \overset{i.i.d.}{\sim} P_c$.

**Assumption 3** (A2 score stationarity across chromosomes)**.** *A2-marginal: score law is chrom-invariant. A2-class: invariant conditional on* $Y = k$*. A2-cell: invariant conditional on* $Y = k$ *and* $\hat{\sigma}$*-bin. Hierarchy: A2-cell* $\Rightarrow$ *A2-class* $\Rightarrow$ *A2-marginal.*

Empirically, the cell-level KS audit (App. B, Tab. 4) finds approximate stationarity on Complex ($10.4\%$ rejection across $125$ tests, near the $5\%$ null) and pronounced cell-level violations on Mendelian ($46.2\%$ rejection across $52$ tests, max KS $0.434$); T3′ is the operational certificate on Mendelian (§5.1), T3 exact suffices on Complex.

**Assumption 4** (A-SL Score-Lipschitz regularity)**.** *For each class* $k$*,* $\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| \leq L_F|\sigma_1 - \sigma_2|$ *for all* $\sigma_1, \sigma_2 \in [\hat{\sigma}_{\min}, \hat{\sigma}_{\max}]$.

4 is used only by T5 (§5.2) to derive the optimal bin count $K^\star$. Empirical audit of 4 via per-pair Lipschitz ratios on $\hat{\sigma}$-bin pairs (App. B, Fig. 6) gives noise-adjusted upper envelope $L_F^{(95\%)} = 13.8$ (Mendelian) and $8.1$ (Complex); LCLS estimator [Huang et al., 2024] gives a tighter typical-slope estimate $\hat{L}_F = 2.97$ / $4.51$.

## 3.2 Score and prediction set

**Aggregator** $\hat{p}$ (HistGradientBoosting depth 2, class-balanced) and **heteroscedastic head** $\hat{\sigma}$ (same backbone, Gaussian NLL on chrom-LOO OOF residuals $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$). **Nonconformity score** $s(x, y) := |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$, $\epsilon = 10^{-6}$. **Mondrian prediction set**: partition $\mathcal{D}^{(-c^\star)}$ into $2K$ cells via $(Y_i, \mathrm{bin}_K(\hat{\sigma}(X_i)))$ where $\mathrm{bin}_K$ uses calibration-fold $\hat{\sigma}$-quantiles. Per-cell threshold:

$$\hat{q}_{k,b} := \mathrm{Quantile}\big(\{s(X_i, Y_i) : Y_i = k, \mathrm{bin}_K(\hat{\sigma}(X_i)) = b\},\ \lceil (n_{k,b} + 1)(1 - \alpha)\rceil / n_{k,b}\big). \quad (1)$$

The Mondrian prediction set:

$$\mathcal{C}_\alpha(x) := \{k \in \{0,1\} : |k - \hat{p}(x)| \leq \hat{q}_{k,\mathrm{bin}_K(\hat{\sigma}(x))} \cdot (\hat{\sigma}(x) + \epsilon)\}. \quad (2)$$

For cells with $n_{k,b} < \lceil 1/\alpha \rceil$ we fall back to the pooled class-conditional threshold $\hat{q}_k$ (T2).

# 4 Method: HCCP

**Pipeline.** HCCP is a three-stage post-hoc framework on top of any probabilistic classifier: (1) aggregator $\hat{p}$, (2) heteroscedastic head $\hat{\sigma}$, (3) Mondrian-($y \times \hat{\sigma}$-bin) conformal calibration. Each stage is decoupled, so HCCP accepts any $(\hat{p}, \hat{\sigma})$ pair as a drop-in. Algorithm 1 states the full pipeline; per-cell small-cell fallback to the pooled T2 threshold triggers when $n_{k,b} < \lceil 1/\alpha \rceil$.

**Why the joint ($y \times \hat{\sigma}$-bin) partition.** Pure class-conditional calibration (Mondrian on $y$ alone) [Vovk et al., 2003] fixes the imbalance pathology of split CP but leaves within-class local coverage unaddressed: our ablation reaches $\mathrm{cov}_{|Y=1} = 0.896$ on Complex but the hardest $\hat{\sigma}$-decile has coverage $0.454$. Stratifying by $\hat{\sigma}$-bin alone re-introduces minority-class pathology because $\hat{\sigma}$-distribution is dominated by negatives. Among Mondrian partitions over $\{y, b(x)\}$, the product partition $(y \times b)$ is the unique refinement that admits both T2 (which requires conditioning on $y$) and T3 (which requires conditioning on $b$); the two strict coarsenings each violate one of the two coverage targets.

**Algorithm 1** HCCP — Heteroscedastic Class-Conditional Conformal Prediction

**Require:** Data $\mathcal{D}$, test chrom $c^\star$, miscoverage $\alpha$, bin count $K$

$\mathcal{D}_{\text{tr}} \leftarrow \{i : C_i \neq c^\star\}$; fit $\hat{p}$ on $\mathcal{D}_{\text{tr}}$ with chrom-LOO OOF predictions $\{\hat{p}^{(-C_i)}(X_i)\}$
Fit $\hat{\sigma}$ on $\mathcal{D}_{\text{tr}}$ with targets $\{|Y_i - \hat{p}^{(-C_i)}(X_i)|\}$ under Gaussian NLL
Compute $\hat{\sigma}$-bin edges via calibration-fold quantiles; calibration scores $s_i = |Y_i - \hat{p}(X_i)|/(\hat{\sigma}(X_i) + \epsilon)$
Partition $\mathcal{D}_{\text{tr}}$ by $(Y_i, \text{bin}_K(\hat{\sigma}(X_i)))$; compute $\hat{q}_{k,b}$ via Eq. (1)
**for** cells with $n_{k,b} < \lceil 1/\alpha \rceil$ **do**
    $\hat{q}_{k,b} \leftarrow \hat{q}_k$ ▷ T2 pooled class-conditional fallback
**end for**
**for** $x \in \mathcal{D}_{\text{te}}$ **do**
    $\mathcal{C}_\alpha(x) \leftarrow \{k \in \{0,1\} : |k - \hat{p}(x)| \leq \hat{q}_{k,b(x)} \cdot \hat{\sigma}(x)\}$
**end for**
**return** $\{(x, \mathcal{C}_\alpha(x))\}$

**$K$ selection: nested chrom-LOO.** We select $\hat{K}(c_{\text{outer}})$ per outer fold via *proper nested chrom-LOO cross-validation*, ensuring $\hat{K}$ is a deterministic function of the outer calibration set alone (CP exchangeability-preserving). For each outer fold, run inner chrom-LOO and pick $\hat{K}(c_{\text{outer}}) = \text{argmin}_K \, \overline{G}_{\text{inner}}(K)$ where the inner-fold worst-cell gap is measured on a fixed-$K_{\text{eval}}$ sigma-bin partition (not on each candidate $K$ itself, which would artificially favour smaller $K$ via fewer cells). $K_{\text{eval}}$ is set per dataset to keep $\geq 100$ minority-class samples per cell: $K_{\text{eval}} = 3$ on Mendelian, $K_{\text{eval}} = 5$ on Complex. Modal $\hat{K}(c_{\text{outer}})$ is 5 on Complex and 5–8 on Mendelian (App. C).

# 5 Theory

This section states HCCP's four main-text theorems. Auxiliary lemmas (T1/T2 corollaries; T1′/T2′/T4 robust counterparts via Barber et al. [2023] Thm. 2; T3-loc feature-ball extension; T3.b $\hat{\sigma}$-perturbation robustness) are stated and proved in App. A.

## 5.1 Bin-conditional coverage (T3 / T3′)

**Theorem 1** (T3 bin-conditional coverage, exact)**.** *Under 2 and A2-cell, for each cell with $n_{k,b} \geq \lceil 1/\alpha \rceil$ and continuous $F_{kb}$, $1 - \alpha \leq P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \leq 1 - \alpha + 1/(n_{k,b} + 1)$.*

**Positioning vs. Barber et al. [2020].** We do not claim pointwise conditional coverage $P(Y \in \mathcal{C}_\alpha(X) \mid X = x) \geq 1 - \alpha$, ruled out in finite samples without structural assumptions. T3 conditions on the $(y, b)$-partition event; the $\hat{\sigma}$-bins are the coarsest practically useful strengthening of standard class-conditional CP that still carries finite-sample bounds.

**Theorem 2** (T3′ bin-conditional coverage, robust)**.** *Let $\bar{\delta}_{\text{TV}}^{(k,b)}$ be the average TV distance between calibration-fold and test-fold conditional score distributions in cell $(k, b)$. Under 2 alone (without A2-cell), for each cell with $n_{k,b} \geq \lceil 1/\alpha \rceil$, $P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \geq (1 - \alpha) - \bar{\delta}_{\text{TV}}^{(k,b)}$.*

T3′ is the operative *certificate* on Mendelian rather than a tight prediction: empirical $\bar{\delta}_{\text{TV}}^{(k,b)} \leq 0.31$ on the worst cell yields a coverage lower bound of $0.59$, which is loose by construction (uniform over all distributions satisfying 2 alone), while the observed per-chromosome coverage gap remains $\leq 0.100$ throughout (chr6 at $0.800$; full per-chrom breakdown in App. C.4). T3′ supplies the only uniform finite-sample lower bound on coverage when T3's exactness assumption (A2-cell) is empirically rejected. Our per-chromosome KS audit (App. B) shows the rejection rate is approximately $K$-invariant across $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$ (40–46% on Mendelian), so T3′ supplies the certificate at every partition granularity.

## 5.2 Optimal bin count and within-class tight rate (T5.1, T5.2)

This is the principal theoretical contribution. T5.1 derives the optimal $K$ by formalising the bias-variance tradeoff and T5.2 proves a finite-sample lower bound that is *tight within the equi-bin*

*Mondrian-$K$ family* (i.e., a within-class tightness statement, not a procedure-class minimax). The pair establishes that the $O(n^{-1/2})$ rate is tight on this class, dimension-free, and depends explicitly on $\pi_{\min}$ via the $\pi_{\min}^{-1/2}$ constant; concurrent split-CP rate analyses (e.g., Yao et al. [2026], also $O(n^{-1/2})$ dim-free, no Mondrian / $\pi_{\min}$ / lower bound) and other axes (e.g., self-calibrating CP [van der Laan, 2024] on the $\hat{\mu}$-axis with $O(n^{-2/3})$) are not subsumed and do not subsume ours.

Under 4, the worst-cell coverage gap decomposes as

$$G(K) := \max_{k,b} |\mathrm{cov}_{k,b} - (1-\alpha)| \leq \underbrace{\frac{L_F R}{K}}_{\text{stationarity bias}} + \underbrace{\frac{K}{\pi_{\min} n}}_{\text{finite-sample variance}}, \tag{3}$$

where $R = \hat{\sigma}_{\max} - \hat{\sigma}_{\min}$ and $\pi_{\min} = \min(\pi, 1-\pi)$.

**Theorem 3** (T5.1 oracle bin count)**.** *Under 2 and 4 with equiprobable $\hat{\sigma}$-bins, $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ minimises Eq.* (3)*, with $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$.*

**Theorem 4** (T5.2 within-class tight finite-sample lower bound on the equi-bin Mondrian-$K$ class)**.** *Restrict attention to conformal predictors using equiprobable $\hat{\sigma}$-bin Mondrian partitions on the calibration fold (the class HCCP belongs to). There exists a distribution satisfying 2 and 4 such that, for any $K$ within this class, $\min_K G(K) = \Omega\big(\sqrt{L_F R/(\pi_{\min} n)}\big)$. Hence the $O(n^{-1/2})$ rate of T5.1 is tight on this class, and the $\pi_{\min}^{-1/2}$ constant is not improvable without strengthening assumptions or moving outside the class.* This is a within-class tightness statement: it does not rule out lower bounds in richer procedure classes (data-dependent partitions, weighted CP, RLCP, SC-CP); it certifies that the equi-bin Mondrian-$K$ family — within which HCCP is instantiated — admits no asymptotically faster $K$-selection.

**Proof sketch.** For T5.1: $g'(K) = -L_F R/K^2 + 1/(\pi_{\min} n) = 0$ gives $K^\star = \sqrt{L_F R \pi_{\min} n}$. For T5.2: a linear-shift family $F_{k,\sigma}(s) = \Phi(s - L_F \sigma)$ matches both bias and variance lower bounds. Full proofs in App. A.11.

**Practical $\hat{K}$ via nested chrom-LOO.** *T5.1 is descriptive of the asymptotic rate, not prescriptive of $K$ at finite $n$.* The plug-in LCLS $\hat{L}_F$ [Huang et al., 2024] gives asymptotic $K^\star = 22$ (Mendelian) / 50 (Complex), but the per-cell minority floor $n_{k,b} \geq \lceil 1/\alpha \rceil$ binds well before $K^\star$ at TraitGym scale ($n \leq 11{,}400$); modal nested-CV $\hat{K}(c_{\text{outer}}) = 5$ Complex / $\{5, 8\}$ Mendelian (3–10$\times$ smaller), reflecting the finite-sample fallback regime (App. C). The synthetic $n$-sweep (App. F) verifies the rate: log-log slope $-0.50 \pm 0.05$ across feature dimensions $d \in \{8, 32, 128\}$, confirming dimension-free. Practical recommendation: use nested chrom-LOO CV with K-grid $\{2, 3, 5, 8, 10\}$ for finite-sample $K$-selection; T5.1's $K^\star$ guides only the asymptotic regime.

### 5.3 Summary of guarantees

Table 1: HCCP coverage guarantees at a glance. Auxiliary lemmas (T1, T2, T1′, T2′, T3-loc, T3.b, T4) live in App. A.

| Result | Assumptions | Guarantee |
|---|---|---|
| T3 (Thm. 1) | A1′ + A2-cell | $\mathrm{cov}_{k,b} \in [1-\alpha, 1-\alpha+1/(n_{k,b}+1)]$ |
| T3′ (Thm. 2) | A1′ only | $\mathrm{cov}_{k,b} \geq (1-\alpha) - \bar{\delta}_{\mathrm{TV}}^{(k,b)}$ |
| T5.1 (Thm. 3) | A1′ + 4 | $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$, $K^\star = O(\sqrt{n})$ |
| T5.2 (Thm. 4) | equi-bin Mondrian-$K$ | $\min_K G(K) = \Omega(\sqrt{L_F R/(\pi_{\min} n)})$ |

## 6 Experiments

**Setup.** TraitGym Mendelian matched_9 ($n = 3{,}380$, 113 OMIM) and Complex matched_9 ($n = 11{,}400$, 29 traits); chrom-LOO with test chroms $\{17{-}22, X\}$ (TraitGym convention). Features: CADD [Kircher et al., 2014], Borzoi [Linder et al., 2025], GPN-MSA [Benegas et al., 2023]. AUPRC chrom-weighted, $\alpha = 0.10$. HCCP partition $K = \hat{K}(c_{\text{outer}})$ via nested chrom-LOO; reporting metric on fixed $K_{\text{eval}}$ (Mendelian $K_{\text{eval}} = 3$, Complex $K_{\text{eval}} = 5$, set per dataset to keep $\geq 100$

Table 2: TraitGym chrom-LOO. HCCP at $K = 5$ for direct comparability with the split-CP and class-conditional baselines (the recommended Mondrian operating point $K_{\text{eval}} = 3/5$ for Mendelian / Complex is reported in Tab. 8). AUPRC chrom-weighted; $\hat{\sigma}$-bin gap here is the **max-over-$\hat{\sigma}$-bins marginal coverage deviation from** $0.90$ (averaged across class within each bin); the cell-level worst-$(k, b)$ gap that appears in T5's bound is reported in Tab. 8.

| Dataset | Aggregator | Calibration | AUPRC ↑ | Cov\|$Y$=1 | $\hat{\sigma}$-bin gap ↓ | Singleton |
|---|---|---|---|---|---|---|
| Mendelian | LogReg [Benegas et al., 2025] | Split CP | 0.752 | 0.89 | 0.54 | 91% |
| | GBM (ours) | Class-cond. (T2 only) | **0.900** | 0.90 | 0.46 | 92% |
| | GBM | DEGU-lite + split | 0.901 | 0.88 | 0.43 | 85% |
| | GBM | **HCCP (ours)** | **0.900** | $0.894_{\pm 0.015}$ | $\mathbf{0.020}_{\pm 0.012}$ | $0.776_{\pm 0.040}$ |
| Complex | LogReg | Split CP | 0.332 | 0.88 | 0.43 | 23% |
| | GBM | Class-cond. (T2 only) | 0.353 | 0.90 | 0.51 | 31% |
| | GBM | DEGU-lite + split | 0.353 | 0.86 | 0.43 | 20% |
| | GBM | **HCCP (ours)** | 0.353 | $0.904_{\pm 0.004}$ | $\mathbf{0.004}_{\pm 0.002}$ | $0.228_{\pm 0.019}$ |

minority-class samples per cell). **Coverage and gap statistics throughout the H2H comparison are $B = 200$ chromosome-level bootstrap mean $\pm$ std for HCCP and all matched-$(\hat{p}, \hat{\sigma})$ baselines (Tab. 8, Tab. 9).** The aggregator backbone comparison in Tab. 2 reports single-seed point estimates for the non-HCCP backbones (different aggregator pipelines); HCCP-side bootstrap CIs are unchanged from Tab. 8 (full setup in App. B).

### 6.1 Main results

The GBM aggregator alone reaches SOTA Mendelian AUPRC $0.900$, a $+14.8$ pp improvement over the published LogReg baseline [Benegas et al., 2025]. Class-conditional calibration fixes the minority-class pathology of split CP ($\text{cov}_{|Y=1}$ $0.54 \to 0.90$ on Complex) but leaves the local gap at $0.46$–$0.51$. HCCP's joint $(y \times \hat{\sigma}\text{-bin})$ Mondrian reduces the marginal $\hat{\sigma}$-bin gap by a further 5–25$\times$, to within one standard error of the finite-sample $1/(n_{k,b} + 1)$ floor. The corresponding cell-level worst-$(k, b)$ gap (the object that appears in T5's bound) is larger by construction: $0.173 \pm 0.192$ Mendelian and $0.060 \pm 0.031$ Complex at the recommended $K_{\text{eval}}$ (Tab. 8); the two metrics differ by $\sim 8\times$ on Mendelian because minority-cell undercoverage drives the cell-level worst, which the marginal-bin metric averages out.

### 6.2 Head-to-head against contemporary CP

We compare HCCP (B4) against six contemporary CP variants at matched $(\hat{p}, \hat{\sigma})$: split (B1), $\hat{\sigma}$-Mondrian (B2, [Boström and Johansson, 2020]), class-Mondrian (B3, [Sadinle et al., 2019]), RLCP [Hore and Barber, 2025], weighted CP [Tibshirani et al., 2019], and SC-CP [van der Laan, 2024]. **At the recommended operating point ($K_{\text{eval}} = 3$ Mendelian, $5$ Complex; per-cell minority $\geq 100$), HCCP is the only single-fold partition we evaluated that holds $\text{cov}_{|Y=1} \geq 0.85$ and $\hat{\sigma}$-bin gap $\leq 0.20$ on both datasets** (Tab. 8, Tab. 9; all rows $B = 200$ chrom-bootstrap mean $\pm$ std). HCCP tightens the bin-local gap by $13.7\times$ on Complex / $2.3\times$ on Mendelian over the on-axis baseline B2 $\hat{\sigma}$-Mondrian (HCCP $0.060/0.173$ vs. B2 $0.827/0.401$, bootstrap means), and $13.4\times/2.4\times$ over the off-axis B3 class-Mondrian. The two ratios coincide because B2 $\hat{\sigma}$-stratification still loses minority coverage ($\text{cov}_{|Y=1} = 0.624$ on Complex), so its $\hat{\sigma}$-bin gap is dominated by minority-cell undercoverage: the joint $(y \times \hat{\sigma})$ product binds, not the $\hat{\sigma}$ axis alone. The Mendelian advantage narrows under weighted CP (HCCP $0.173 \pm 0.192$ vs. wCP $0.252 \pm 0.153$, $1.5\times$); a $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$ sweep (App. C.3) shows HCCP outperforms all baselines across the full range on Complex (3–34$\times$) but reverses on Mendelian at $K_{\text{eval}} \geq 5$ (per-cell minority $< 70$). The reversal traces directly to Eq. (3): below the per-cell variance-stable floor, the $K/(\pi_{\min} n)$ variance term overtakes the $L_F R/K$ Lipschitz bias gain, and the smoother reweighting of weighted CP avoids the hard-bin variance penalty. This is a regime boundary, not a model-class limitation. The same calibration on the published LogReg aggregator yields essentially identical $\hat{\sigma}$-bin gap (App. C.2), decoupling the conformal-side gain from base-predictor accuracy.

**Imbalance scaling validates T5's $\pi_{\min}$ dependence.** Subsampling Complex to $\pi_+ \in \{0.05, 0.10, 0.20, 0.30, 0.50\}$, B1 and B2 minority-coverage degrades monotonically as $\pi_+ \to 0$

($0.587 \to 0.810$ for B1; $0.650 \to 0.878$ for B2), while HCCP and class-Mondrian hold at the target across the full sweep (App. C, Tab. 10) — the empirical realisation of the $\pi_{\min}^{-1/2}$ factor in T5's bound.

### 6.3 Three-axis OOD + cross-domain stress test

We evaluate HCCP on three orthogonal distributional axes (chrom-LOO, trait-LOO, cross-dataset) and a cross-domain benchmark (ProteinGym). *Trait-LOO* reaches the finite-sample $1/(n_{k,b}+1)$ floor on both datasets ($\hat{\sigma}$-bin gap $\leq 0.004$), with a single exception: the Skeletal/connective OMIM cluster ($n_{\text{tot}} = 470$) shows $\text{cov}_{|Y=1} = 0.745$ ($1/3$ Mendelian clusters fall below the $0.85$ minority-coverage target). *Chrom-LOO* on the SOTA feature set: per-chrom |coverage gap| $\leq 0.100$ Mendelian (chr6 at $0.800$) and $\leq 0.023$ Complex (Tab. 13). *Cross-dataset* C→M holds (gap $0.035$); the reverse M→C direction is a joint-failure case study (§7, R3). *ProteinGym cross-domain* [Notin et al., 2023]: across 50 assay-LOO held-out proteins, HCCP attains marginal coverage $0.887$, within $\pm 0.02$ of target on $27/50$ assays and within $\pm 0.05$ on $44/50$. The 6 outlier assays correlate with low base-predictor AUPRC (Spearman $\rho = 0.46$, $p = 0.001$), confirming coverage failures track base-predictor weakness rather than conformal failure (full diagnosis in App. E).

### 6.4 Ablations

Ablations on $K$-sweep (T5 rate validation), $\hat{\sigma}$ source (supervised vs. DEGU-lite ensemble), $\hat{\sigma}$ target function (residual vs. log-variance), nonconformity score, class-conditional on/off, AlphaGenome [Avsec et al., 2026] feature extension, and synthetic $n$-sweep (verifying the $O(n^{-1/2})$ rate of Theorem 3) are reported in App. C and App. F. The synthetic sweep is the direct test of T5.1's dimension-free claim: log-log slope $-0.50 \pm 0.05$ across $d \in \{8, 32, 128\}$ matches the $\pi_{\min}^{-1/2}$ predicted constant. The DEGU-lite ablation [Zhou et al., 2026] swaps our supervised residual $\hat{\sigma}$ for an ensemble-distilled Gaussian-uncertainty $\hat{\sigma}$ at matched GBM backbone and matched HCCP-style Mondrian calibration; this isolates the choice of variance source within HCCP and is *not* a head-to-head against DEGU's published Keras CNN MPRA pipeline. With this caveat, the supervised residual head outperforms the distilled ensemble head by $4.3\times$ on Complex worst-cell coverage, identifying where aleatoric-aware $\hat{\sigma}$ binds (polygenic complex traits) and where ensemble disagreement suffices (monogenic Mendelian).

## 7 Discussion

**Clinical implications.** HCCP prediction sets are not a replacement for expert review; they are a calibrated abstention signal. A singleton set $\mathcal{C}_\alpha(x) = \{k\}$ is a confident call; a set containing both labels or neither is an explicit abstention. On TraitGym Mendelian, HCCP produces singletons for $\approx 78\%$ of variants at $\alpha = 0.10$ ($K_{\text{eval}} = 5$, Tab. 2; cf. $0.827$ at $K_{\text{eval}} = 3$ in Tab. 9). Operating-point selection by miscoverage level is dataset-specific: $\alpha = 0.10$ for high-stakes rare-disease classification, $\alpha = 0.20$ for population-screening triage of complex traits where the abstention burden is otherwise prohibitive (cf. Adegoke et al. [2025] report $41\%$ abstention at $\alpha \approx 0.20$ in a clinical screening setting); App. D reports the singleton-vs-abstention tradeoff across $\alpha \in [0.05, 0.50]$.

**Operational envelope and limitations.** The coverage guarantees of HCCP hold when (a) per-cell minority $\geq 100$, (b) base AUPRC $\gtrsim 0.30$, and (c) per-cell KS-reject $\lesssim 50\%$. 1 (chrom-group exchangeability) is load-bearing for T3; T3′ absorbs A2-cell violations via the chrom-TV term on Mendelian ($46.2\%$ reject, $K$-invariant in $[40, 46]\%$ across $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$; App. B); 4 is required only for T5. We report four labelled out-of-envelope regimes as case studies, not coverage claims: **(R1)** Mendelian $K_{\text{eval}} \geq 5$ violates (a) at per-cell minority $< 70$ (§6.2; reversal *predicted* by Eq. (3)); **(R2)** the Skeletal/connective OMIM cluster ($n = 470$, $1/3$ Mendelian clusters) violates (a) with $\text{cov}_{|Y=1} = 0.745$ (§6.3); **(R3)** the M→C cross-dataset direction jointly violates (b) (AUPRC $0.205$ vs. ID $0.353$) and (c) (ClinVar vs. GWAS catalog selection [Tibshirani et al., 2019]), with marginal coverage $0.738$ ($0.004$ below the Barber 2023 Thm. 2 proxy bound of $0.742$); the forward C→M and ProteinGym cross-domain violate neither precondition and HCCP holds; **(R4)** ProteinGym $6/50$ outliers violate (b) (CD19 AUPRC $0.24$; App. E). **Scope**: (i) marginal-score TV proxy underestimates swap-TV by $\approx 0.036$ on M→C (App. B); (ii) oracle $K^\star$ is large-sample, the binding constraint at $n \leq 11{,}400$ is the per-cell floor $n_{k,b} \geq \lceil 1/\alpha \rceil$, modal nested-CV $\hat{K} \in \{5, 8\}$ Mendelian / 5 Complex (App. C); (iii) DEGU CNN end-to-end port is out of scope (200 bp DNA

windows); we ablate the $\hat{\sigma}$-mechanism only at matched GBM backbone; (iv) per-assay $\hat{\sigma}$ adaptation for R4 outliers remains future work.

**Broader impact.** VEP systems inform consequential clinical decisions (cascade testing, reproductive choices). HCCP provides calibrated abstention but does not guarantee clinical safety: the $\approx 78\%$ singletons on Mendelian still carry a $10\%$ finite-sample miscoverage floor, and abstention must be paired with expert review to translate coverage into clinical utility. Population-stratified deployment requires re-calibration on ancestry-conditional folds (a fourth Mondrian axis); TraitGym is built from ClinVar and the GWAS catalog and over-represents European-ancestry individuals. The ACMG/AMP consensus [Richards et al., 2015] treats computational predictions as PP3/BP4 supporting evidence, not stand-alone classification; HCCP singletons should be integrated into multi-evidence frameworks, not deployed as diagnostic.

# 8 Conclusion

We introduced HCCP, a post-hoc framework for variant effect prediction that pairs a learned heteroscedastic head $\hat{\sigma}$ with Mondrian-$(y \times \hat{\sigma}\text{-bin})$ conformal calibration. Three contributions: (1) the first finite-sample $O(n^{-1/2})$ rate for joint class-conditional + heteroscedastic-local CP, with explicit $\pi_{\min}^{-1/2}$ dependence and a within-class matching $\Omega(n^{-1/2})$ lower bound on the equi-bin Mondrian-$K$ family (Theorems 3, 4; not a procedure-class minimax: SC-CP attains $O(n^{-2/3})$ on a different axis); (2) the HCCP framework, with T3 exact bin-conditional coverage and a Barber-2023 robustness corollary T3′ that is non-vacuous on Complex but loose-by-construction on Mendelian (KS rejects A2-cell exchangeability at $46.2\%$); (3) empirical validation on TraitGym + ProteinGym + synthetic, decisively dominant on Complex (HCCP $\hat{\sigma}$-bin gap $0.060 \pm 0.031$ vs. $\geq 0.805$ across all six matched-$(\hat{p}, \hat{\sigma})$ baselines we evaluated; $B = 200$ chrom-bootstrap CIs disjoint), with a more contested signal on Mendelian (HCCP $\pm 1\sigma$ band overlaps weighted CP at $K_{\text{eval}} = 3$, the only operating point with per-cell minority $\geq 100$; the regime boundary at $K_{\text{eval}} \geq 5$ is predicted by Eq. 3's bias-variance decomposition, not a model-class limitation). The conformal-side gain is decoupled from the choice of base predictor (HCCP-on-LogReg gives an essentially identical $\hat{\sigma}$-bin gap, App. C.2); a synthetic $n$-sweep verifies the dimension-free $O(n^{-1/2})$ rate (slope $-0.50 \pm 0.05$). Natural extensions: end-to-end joint training of $(\hat{p}, \hat{\sigma})$, richer base predictors (AlphaGenome [Avsec et al., 2026]), and population-stratified Mondrian for ancestry-conditional coverage.

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# A Full proofs of T1–T5

## A.1 Assumptions hierarchy

We collect in one place the assumptions invoked across §5. The chain of A2 variants is:

$$\underbrace{\text{A2-cell}}_{\text{T3}} \implies \underbrace{\text{A2-class}}_{\text{T2}} \implies \underbrace{\text{A2-marginal}}_{\text{T1}},$$

where each level is obtained from the stronger one by marginalizing over a partition variable.

- **A1** (chrom-group exchangeability, 1): conditional on $C = c$, $\{(X_i, Y_i) : C_i = c\}$ are exchangeable with law $P_c$; different chromosomes are mutually independent.
- **A1′** (chrom-wise i.i.d., 2): the stronger version in which $\{(X_i, Y_i) : C_i = c\}$ are i.i.d. $P_c$.
- **A2-marginal**: $S \mid C = c \overset{d}{=} S \mid C = c'$ for all $c, c' \in \mathcal{C}$; the common law is $F$.

- **A2-class**: $S \mid (Y = k,\, C = c) \overset{d}{=} S \mid (Y = k,\, C = c')$ for all $c, c',\, k \in \{0, 1\}$; common law $F_k$.
- **A2-cell**: $S \mid (Y = k,\, b(X) = b,\, C = c) \overset{d}{=} S \mid (Y = k,\, b(X) = b,\, C = c')$ for all $c, c',\, k,\, b$; common law $F_{kb}$.
- **A3-loc** (local stationarity, used in T3-loc): for every $(x_0, r)$ satisfying the T3-loc resolution condition, $s(X, k) \mid (Y = k,\, b(X) = b,\, X \in B_\ell(x_0, r))$ is distributed as $F_{kb}$.
- **B1** ($\hat{\sigma}$ relative error, used in T3.b): $\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \le \eta$ for some $\eta \in (0, 1)$.
- **A-SL** (score-Lipschitz regularity, 4, used in T5): $\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| \le L_F \cdot |\sigma_1 - \sigma_2|$ for all $\sigma_1, \sigma_2 \in [\hat{\sigma}_{\min}, \hat{\sigma}_{\max}]$ and $k \in \{0, 1\}$.

**Empirical validation of A2.** The within-cell KS audit (Tab. 4) ran 177 cell-level two-sample KS tests across both datasets ($K_{\text{eval}} = 5$ bins $\times$ 2 classes $\times$ test chromosomes with non-empty cells; 52 tests on Mendelian, 125 on Complex). The rejection rates at nominal $p < 0.05$ are $46.2\%$ (Mendelian, max KS $0.434$) and $10.4\%$ (Complex, max KS $0.315$). Since $d_{\text{TV}} \le d_{\text{KS}}/\sqrt{2}$ does *not* hold in general — only $d_{\text{TV}} \le d_{\text{KS}}$ for univariate $F$, with the tighter $\sqrt{2}$ bound under additional regularity — we use the conservative envelope $\bar{\delta}_{\text{TV}}^{(k,b)} \le d_{\text{KS}}$, giving worst-case $\bar{\delta}_{\text{TV}}^{(k,b)} \le 0.31$ on the binding Mendelian cell (chr9, chr7, chr3 pathogenic-rich at the highest-$\hat{\sigma}$ bin). A2-cell holds approximately on Complex; T3$'$ supplies the certificate on Mendelian.

### A.2 Auxiliary lemmas

We state four standard lemmas used in the proofs below.

**Lemma 1** (Rank uniformity for exchangeable sequences)**.** *Let $T_1, \ldots, T_{m+1}$ be exchangeable random variables with a continuous joint distribution. Then the rank of $T_{m+1}$ among $(T_1, \ldots, T_{m+1})$ is uniformly distributed on $\{1, \ldots, m+1\}$.*

*Proof.* By exchangeability, for any permutation $\pi$ of $\{1, \ldots, m+1\}$, the joint law of $(T_{\pi(1)}, \ldots, T_{\pi(m+1)})$ equals that of $(T_1, \ldots, T_{m+1})$. Continuity implies $P(T_i = T_j) = 0$ for $i \ne j$, so the ranking is almost surely unique. For each $r \in \{1, \ldots, m+1\}$, the event $\{\text{rank}(T_{m+1}) = r\}$ can be mapped to the event $\{\text{rank}(T_{\pi(m+1)}) = r\}$ by a suitable permutation that places the element currently at rank $r$ in position $m+1$, so all rank values are equiprobable. □

**Lemma 2** (Exchangeable CP quantile bound)**.** *Under the conditions of Lemma 1, let $\hat{q} = \text{Quantile}(\{T_1, \ldots, T_m\},\, \lceil (m+1)(1-\alpha) \rceil / m)$ be the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of the first $m$ values. Then*

$$P(T_{m+1} \le \hat{q}) \;=\; \frac{\lceil (m+1)(1-\alpha) \rceil}{m+1} \;\in\; \left[1 - \alpha,\; 1 - \alpha + \frac{1}{m+1}\right].$$

*Proof.* By Lemma 1, $T_{m+1}$ is at rank $r$ with probability $1/(m+1)$ for each $r$. The event $\{T_{m+1} \le \hat{q}\}$ is exactly $\{\text{rank}(T_{m+1}) \le \lceil (m+1)(1-\alpha) \rceil\}$ (since $\hat{q}$ is the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of $(T_1, \ldots, T_m)$, which is also the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of the augmented set when $T_{m+1}$ has rank at most that value). Thus

$$P(T_{m+1} \le \hat{q}) = \frac{\lceil (m+1)(1-\alpha) \rceil}{m+1}.$$

Since $\lceil (m+1)(1-\alpha) \rceil \in [(m+1)(1-\alpha),\, (m+1)(1-\alpha) + 1)$, dividing by $m+1$ gives the stated bounds. □

**Lemma 3** (Monotone score-rescaling invariance)**.** *Let $g : \mathbb{R}_+ \to \mathbb{R}_+$ be a strictly increasing function. If $\hat{q}$ is the conformal quantile computed from scores $\{S_i\}$ and $\hat{q}' = g(\hat{q})$ is the quantile computed from rescaled scores $\{g(S_i)\}$, then for any test score $S_{\text{test}}$,*

$$S_{\text{test}} \le \hat{q} \quad \Longleftrightarrow \quad g(S_{\text{test}}) \le \hat{q}'.$$

*Proof.* The quantile index $\lceil (m+1)(1-\alpha) \rceil$ is the same for both sets. Since $g$ is strictly increasing, it preserves the ordering of scores. Thus the $j$-th order statistic of $\{g(S_i)\}$ equals $g$ applied to the $j$-th order statistic of $\{S_i\}$, giving $\hat{q}' = g(\hat{q})$. The equivalence follows from $g$ being order-preserving. □

**Lemma 4** (Bin-boundary mass bound)**.** *For equiprobable bins with $K$ bins and minimum bin width $\Delta$, the probability mass of $\sigma^\star(X)$ within distance $\delta$ of any bin edge is at most $2K\delta/\Delta$.*

*Proof.* Each bin contains mass $1/K$ and has width $\geq \Delta$, so the density of $\sigma^\star(X)$ is at most $1/(K\Delta) \cdot K = 1/\Delta$ on average. There are $K-1$ internal bin edges, and each has a neighborhood of width $2\delta$ that captures mass at most $2\delta/\Delta$. Summing over all edges gives $2(K-1)\delta/\Delta \leq 2K\delta/\Delta$. □

### A.3 Lemma 5: T1 marginal coverage

**Lemma 5** (T1 marginal coverage — standard split-CP at the marginal partition level)**.** *Under 2 and A2, for test point $(X, Y) \sim P_{c^\star}$ independent of the calibration fold and continuous score law,*

$$1-\alpha \;\leq\; P\big(Y \in \mathcal{C}_\alpha(X)\big) \;\leq\; 1-\alpha+\frac{1}{n_{\min}+1}, \qquad n_{\min} = \min_k n_k.$$

*Proof.* We decompose the marginal coverage via the true label.

*Step 1 (Reduction to per-class coverage).* By the law of total probability:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) = \sum_{k\in\{0,1\}} P(Y_{\text{test}} = k) \cdot P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big).$$

Since $Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})$ reduces to $s(X_{\text{test}}, Y_{\text{test}}) \leq \hat{q}_{Y_{\text{test}}}$ by definition of $\mathcal{C}_\alpha$, we have

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) = P\big(s(X_{\text{test}}, k) \leq \hat{q}_k \mid Y_{\text{test}} = k\big).$$

*Step 2 (Joint exchangeability inside class $k$).* Fix $k$. Under 2, for each calibration chromosome $c \neq c^\star$, $(X_i, Y_i)_{C_i=c}$ are i.i.d. $P_c$. The score $S_i = s(X_i, Y_i)$ is a deterministic function of $(X_i, Y_i)$ via the fixed $(\hat{p}, \hat{\sigma})$ trained on $\mathcal{D}^{(-c^\star)}$. Restricting to $i \in \mathcal{I}_k^{(-c^\star)}$ (i.e., $Y_i = k$) yields i.i.d. samples from a chrom-specific conditional law $F_k^c$. By A2-class (implied by A2-marginal when class prevalence $P(Y = k \mid C)$ is chrom-independent, or invoked directly), $F_k^c = F_k$ for all $c$. Across chromosomes, samples are mutually independent (A1). Hence the full calibration set $\{S_i\}_{i\in\mathcal{I}_k^{(-c^\star)}}$ is i.i.d. $F_k$.

Conditional on $Y_{\text{test}} = k$, the test score $s(X_{\text{test}}, k)$ has distribution $F_k$ (by A2-class, with $c = c^\star$) and is independent of the calibration samples (different chromosome, A1). Thus the augmented sequence $(S_1^{(k)}, \ldots, S_{n_k}^{(k)}, s(X_{\text{test}}, k))$ is i.i.d. $F_k$, hence exchangeable.

*Step 3 (Standard exchangeable CP bound).* By Lemma 2 applied with $m = n_k$ and $T_{m+1} = s(X_{\text{test}}, k)$:

$$P\big(s(X_{\text{test}}, k) \leq \hat{q}_k \mid Y_{\text{test}} = k\big) = \frac{\lceil (n_k+1)(1-\alpha) \rceil}{n_k+1} \in \left[1-\alpha,\; 1-\alpha+\frac{1}{n_k+1}\right].$$

*Step 4 (Assemble).* Substituting into Step 1:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) = \sum_k P(Y_{\text{test}} = k) \cdot P\big(s(X_{\text{test}}, k) \leq \hat{q}_k \mid Y_{\text{test}} = k\big).$$

Each term lies in $[1-\alpha,\, 1-\alpha+1/(n_k+1)]$. The convex combination inherits the range: the lower bound gives $\geq 1-\alpha$; the upper bound gives $\leq 1-\alpha+1/(n_{\min}+1)$ with $n_{\min} = \min_k n_k$. □

### A.4 Lemma 6: T2 class-conditional coverage

**Lemma 6** (T2 class-conditional coverage — standard split-CP at the class partition level)**.** *Under the conditions of Lemma 5 with A2 at class level and $n_k \geq \lceil 1/\alpha \rceil$ for each $k \in \{0, 1\}$,*

$$1-\alpha \;\leq\; P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k\big) \;\leq\; 1-\alpha+\frac{1}{n_k+1}.$$

*Proof.* Fix $k \in \{0, 1\}$ with $n_k \geq \lceil 1/\alpha \rceil$.

*Step 1 (Exchangeability).* Under 2 and A2-class, conditioning on $Y_i = k$ makes $(S_1^{(k)}, \ldots, S_{n_k}^{(k)})$ i.i.d. $F_k$ by the same argument as T1, Step 2. Conditional on $Y_{\text{test}} = k$, $s(X_{\text{test}}, k)$ is independent of calibration and distributed as $F_k$. Thus the joint $(S_1^{(k)}, \ldots, S_{n_k}^{(k)}, s(X_{\text{test}}, k))$ is i.i.d., hence exchangeable.

*Step 2 (Quantile bound).* The prediction-set inclusion event $\{Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\}$ under $Y_{\text{test}} = k$ is $\{s(X_{\text{test}}, k) \leq \hat{q}_k\}$. By Lemma 2 with $m = n_k$:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) = \frac{\lceil (n_k + 1)(1 - \alpha) \rceil}{n_k + 1} \in \left[1 - \alpha,\ 1 - \alpha + \frac{1}{n_k + 1}\right].$$

The sample-size requirement $n_k \geq \lceil 1/\alpha \rceil$ ensures $\lceil (n_k + 1)(1 - \alpha) \rceil \leq n_k$, so $\hat{q}_k < +\infty$ and the prediction set is non-trivial. □

**Remark.** T1 is the law-of-total-probability average of the two T2 per-class bounds, so T2 is strictly stronger. The sample-size requirement is comfortably met on TraitGym: at $\alpha = 0.10$ we need $n_k \geq 10$; the smallest chrom-LOO calibration has $n_1 \geq 30$ (Mendelian) and $n_1 \geq 100$ (Complex).

### A.5 Robust fallback: T1′ and T2′

When A2 fails (e.g., a test chromosome has systematically different variant characteristics), we invoke the following restatement of Barber et al. [2023] Theorem 2 in our notation.

**Barber 2023 Theorem 2 (restated).** Let $Z = (Z_1, \ldots, Z_n, Z_{n+1})$ be the full data ordering with $Z_{n+1}$ as the test point, $R(Z) = (S_1, \ldots, S_{n+1})$ the residual vector, and $R(Z^{(i)})$ the sequence with $Z_i$ and $Z_{n+1}$ swapped. For prespecified weights $w_i \in [0, 1]$ with $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$,

$$P\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \ \geq\ 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \cdot d_{\text{TV}}\big(R(Z),\ R(Z^{(i)})\big).$$

No factor of 2.

*Proof of T1′ and T2′.* Use uniform weights $w_i = 1$, so $\tilde{w}_i = 1/(n + 1)$. The Barber 2023 bound becomes

$$P(\text{coverage}) \ \geq\ 1 - \alpha - \frac{1}{n + 1} \sum_i d_{\text{TV}}\big(R(Z),\ R(Z^{(i)})\big).$$

Under 1 alone (without A2), $R(Z^{(i)})$ differs from $R(Z)$ in the position of one residual: $S_i \leftrightarrow S_{\text{test}}$. If $S_i$ is from chromosome $c$ (so $S_i \sim F_c$) and $S_{\text{test}}$ from $c^\star$ (so $S_{\text{test}} \sim F_{c^\star}$), the swap changes the marginal law of exactly two entries of the residual vector. Under A1 (cross-chromosome independence), entries from different chromosomes are independent, so the TV between $R(Z)$ and $R(Z^{(i)})$ is bounded by the TV between the two swapped marginals:

$$d_{\text{TV}}\big(R(Z),\ R(Z^{(i)})\big) \ \leq\ d_{\text{TV}}(F_c,\ F_{c^\star}).$$

Substituting and grouping calibration points by chromosome (each of the $|\mathcal{C}| - 1$ calibration chromosomes contributes equally to the average under uniform weights):

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) \ \geq\ 1 - \alpha - \bar{\delta}_{\text{TV}}, \qquad \bar{\delta}_{\text{TV}} = \frac{1}{|\mathcal{C}| - 1} \sum_{c \neq c^\star} d_{\text{TV}}(F_c,\ F_{c^\star}).$$

This is T1′.

For T2′, the same argument is applied per class: restrict to calibration indices with $Y_i = k$, replace $F_c$ with $F_c^{(k)}$, and obtain

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) \ \geq\ 1 - \alpha - \bar{\delta}_{\text{TV}}^{(k)}, \qquad \bar{\delta}_{\text{TV}}^{(k)} = \frac{1}{|\mathcal{C}| - 1} \sum_{c \neq c^\star} d_{\text{TV}}\big(F_c^{(k)},\ F_{c^\star}^{(k)}\big).$$

Both bounds carry no factor of 2 (weighted-swap Barber 2023, not jackknife+). □

**Empirical TV estimates.** The cell-level KS audit (App. B.4, 177 tests across both datasets) gives worst-cell $d_{\mathrm{KS}} = 0.434$ on Mendelian (chr9 / chr7 / chr3 highest-$\hat{\sigma}$ pathogenic-rich cell) and $0.315$ on Complex. Using $\bar{\delta}_{\mathrm{TV}}^{(k,b)} \le d_{\mathrm{KS}}$ as the conservative envelope, the cell-level coverage gap is bounded by $\le 0.31$ on Mendelian and $\le 0.05$ on Complex; observed per-chromosome coverage gaps remain $\le 0.100$ throughout (chr6 at $0.800$; full table in App. C.4), well below the certificate.

### A.6 Proof of Theorem 1 (bin-conditional coverage, exact)

*Proof.* The argument is the class-conditional proof of Lemma 6 applied inside a single Mondrian cell $(k, b)$, with A2-cell replacing A2-class. Fix a cell $(k, b)$ with $n_{kb} \ge \lceil 1/\alpha \rceil$.

*Step 1 (Cell restriction).* The event "$Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}})$ given $Y_{\mathrm{test}} = k$" is exactly $\{s(X_{\mathrm{test}}, k) \le \hat{q}_{k,b(X_{\mathrm{test}})}\}$ by the definition of $\mathcal{C}_\alpha$. Conditioning further on $b(X_{\mathrm{test}}) = b$ fixes the threshold to $\hat{q}_{kb}$.

*Step 2 (I.i.d. $F_{kb}$ via A1′ + A2-cell).* Under 2, for each calibration chromosome $c \ne c^\star$, $(X_i, Y_i)_{C_i = c}$ are i.i.d. $P_c$; the score $S_i = s(X_i, Y_i)$ is a deterministic function of $(X_i, Y_i)$ via the fixed $(\hat{p}, \hat{\sigma})$. Restricting to $i \in \mathcal{I}_{kb}^{(-c^\star)}$ (i.e., $Y_i = k$ and $b(X_i) = b$) yields i.i.d. samples from a chromosome-specific conditional law, which by A2-cell equals $F_{kb}$ for every calibration chromosome. Across chromosomes, samples are mutually independent (1). Hence the full calibration set $\{S_i\}_{i \in \mathcal{I}_{kb}^{(-c^\star)}}$ is i.i.d. $F_{kb}$.

*Step 3 (Test-score independence and exchangeability).* Conditional on $Y_{\mathrm{test}} = k$ and $b(X_{\mathrm{test}}) = b$, the test score $s(X_{\mathrm{test}}, k)$ has distribution $F_{kb}$ (by A2-cell, with $c = c^\star$) and is independent of the calibration samples (different chromosome, 1). Thus $(S_1^{(kb)}, \ldots, S_{n_{kb}}^{(kb)}, s(X_{\mathrm{test}}, k))$ is i.i.d. $F_{kb}$, in particular exchangeable.

*Step 4 (CP quantile bound).* By Lemma 2 with $m = n_{kb}$:

$$P\big(s(X_{\mathrm{test}}, k) \le \hat{q}_{kb}\big) = \frac{\lceil (n_{kb}+1)(1-\alpha) \rceil}{n_{kb}+1} \in \left[1-\alpha,\ 1-\alpha+\frac{1}{n_{kb}+1}\right].$$

Combining Steps 1 and 4 yields

$$1-\alpha \ \le\ P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}}) \mid Y_{\mathrm{test}} = k,\, b(X_{\mathrm{test}}) = b\big) \ \le\ 1-\alpha+\frac{1}{n_{kb}+1}. \qquad \square$$

**Local coverage gap bound.** For each cell $(k, b)$,

$$\big|P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) - (1-\alpha)\big| \ \le\ \frac{1}{n_{kb}+1}.$$

### A.7 Lemma 7: T3′ bin-conditional coverage, robust

**Lemma 7** (T3′ bin-conditional coverage, robust)**.** *Under 1 alone (no A2 at any level), for each cell $(k, b)$ with $n_{k,b} \ge \lceil 1/\alpha \rceil$,*

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \ \ge\ 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k,b)}, \qquad \bar{\delta}_{\mathrm{TV}}^{(k,b)} = \frac{1}{|\mathcal{C}|-1}\sum_{c \ne c^\star} d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

*Proof.* Apply Barber et al. [2023] Theorem 2 with calibration weights

$$w_i = \mathbb{1}\{i \in \mathcal{I}_{kb}^{(-c^\star)}\}, \qquad \tilde{w}_i = \frac{w_i}{w_1 + \cdots + w_n + 1}.$$

So $\tilde{w}_i = 1/(n_{kb}+1)$ if $i \in \mathcal{I}_{kb}^{(-c^\star)}$ and 0 otherwise. Barber Theorem 2 gives

$$P(\text{coverage in cell } (k,b)) \ \ge\ 1-\alpha-\sum_{i \in \mathcal{I}_{kb}^{(-c^\star)}} \tilde{w}_i \cdot d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big).$$

If $i$ is from chromosome $c$ (so $Z_i \sim F_c^{(k,b)}$) and $Z_{\mathrm{test}}$ is from $c^\star$ (so $Z_{\mathrm{test}} \sim F_{c^\star}^{(k,b)}$), the swap $Z_i \leftrightarrow Z_{\mathrm{test}}$ changes the marginal law of exactly two entries of the residual vector. Under 1 (cross-chromosome independence), entries from different chromosomes are independent, so the TV between

$R(Z)$ and $R(Z^{(i)})$ is bounded by the TV between the two swapped marginals:

$$d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big) \;\le\; d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

Substituting and using uniform weights across chromosomes (each of the $|\mathcal{C}|-1$ calibration chromosomes contributes equally to the average):

$$P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}}) \mid Y_{\mathrm{test}} = k,\, b(X_{\mathrm{test}}) = b\big) \;\ge\; 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k,b)},$$

where

$$\bar{\delta}_{\mathrm{TV}}^{(k,b)} = \frac{1}{|\mathcal{C}|-1}\sum_{c\neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

The bound has no factor of 2. □

**Relation to T3.** Setting $\bar{\delta}_{\mathrm{TV}}^{(k,b)} = 0$ recovers T3's lower bound exactly. T3′ degrades T3 by precisely the cell-level chromosome-TV quantity that our cell-level KS audit (App. B.4) empirically bounds.

### A.8 Lemma 8: T3-loc feature-ball coverage

**Lemma 8** (T3-loc feature-ball coverage)**.** *Suppose $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ is L-Lipschitz in the $\ell_2$ metric and A3-loc (local score stationarity on $\ell_2$-balls) holds. Let $\Delta = \min_b(e_b - e_{b-1})$ be the minimum $\hat{\sigma}$-bin width. For any $x_0 \in \mathcal{X}$, any radius $r$ with $Lr < \Delta/2$, and any class $k$,*

$$\big|P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0,r),\, Y=k\big) - (1-\alpha)\big| \;\le\; \frac{1}{n_{\min}(x_0,r)+1},$$

*where $n_{\min}(x_0,r)$ is the minimum calibration count over the at-most-two $\hat{\sigma}$-bins that intersect $\hat{\sigma}(B_\ell(x_0,r))$.*

*Proof.* Assume $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ is $L$-Lipschitz and A3-loc holds. Fix $x_0 \in \mathcal{X}$, $r > 0$ with $Lr < \Delta/2$, and class $k$.

*Step 1 (Lipschitz bound on $\hat{\sigma}$-range).* For $x \in B_\ell(x_0,r)$:

$$|\hat{\sigma}(x) - \hat{\sigma}(x_0)| \;\le\; L\|x-x_0\| \;\le\; Lr.$$

So $\hat{\sigma}\big(B_\ell(x_0,r)\big) \subseteq [\hat{\sigma}(x_0) - Lr,\, \hat{\sigma}(x_0) + Lr]$.

*Step 2 (Pigeonhole — at most two bins).* Since $Lr < \Delta/2$, this $\hat{\sigma}$-interval has width $2Lr < \Delta$ and therefore intersects at most two adjacent bins $B_{b_\star}, B_{b_\star+1}$. (An interval of width less than the minimum bin width can straddle at most one bin boundary.)

*Step 3 (Convex decomposition).* Decompose the conditional coverage:

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0,r),\, Y=k\big) = \sum_{b\in\{b_\star,b_\star+1\}} w_b(x_0,r)\cdot P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0,r),\, Y=k,\, b(X)=b\big),$$

where $w_b(x_0,r) := P(b(X)=b \mid X \in B_\ell(x_0,r),\, Y=k)$ are non-negative and sum to 1.

*Step 4 (Within-cell restriction preserves $F_{kb}$).* Inside each cell $(k,b)$, the conformal threshold $\hat{q}_{kb}$ depends only on calibration and is not affected by restricting $X$ to $B_\ell(x_0,r)$. Under A3-loc, $s(X,k) \mid (Y=k,\, b(X)=b,\, X \in B_\ell(x_0,r))$ has the cell score law $F_{kb}$. The T3 argument (App. A.6, Steps 2–4) then applies verbatim inside each cell restricted to the ball:

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0,r),\, Y=k,\, b(X)=b\big) \in \left[1-\alpha,\; 1-\alpha+\frac{1}{n_{kb}+1}\right].$$

*Step 5 (Convex-hull sandwich).* The convex combination over the (at most two) bins $\{b_\star, b_\star+1\}$ lies in

$$\left[1-\alpha,\; 1-\alpha+\frac{1}{n_{\min}(x_0,r)+1}\right],$$

where $n_{\min}(x_0,r) = \min_{b\in\{b_\star,b_\star+1\}} n_{kb}$. □

**Honest limitations.** A3-loc is stronger than A2-cell — it requires local stationarity at the scale $r$, which we cannot fully verify at high feature-space dimension with our $n \leq 11{,}400$. We state T3-loc conditionally on A3-loc. The cell-level KS audit (App. B.4) is the closest finite-sample analog. The Lipschitz constant $L$ of $\hat{\sigma}$ (distinct from the score-CDF Lipschitz $L_F$ of 4) is not estimated in this paper and is deferred to future work; T3-loc is therefore stated as a conditional guarantee, not an operationally-deployed bound. Finally, Barber et al. [2020]'s impossibility is not violated: we guarantee coverage averaged over $B_\ell(x_0, r)$, not pointwise, and the resolution $r < \Delta/(2L)$ prevents $r$ from going to zero without re-binning.

### A.9 Lemma 9: T3.b $\hat{\sigma}$ perturbation

**Lemma 9** (T3.b $\hat{\sigma}$ perturbation)**.** *Let $\sigma^\star$ be an oracle variance function and assume B1 ($\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \leq \eta$). Under A1′ + A2-cell (w.r.t. the fixed bin edges $e_0 < \cdots < e_K$),*

$$\Big| P\big(Y \in \mathcal{C}_\alpha^{(\hat{\sigma})}(X) \mid Y = k\big) - P\big(Y \in \mathcal{C}_\alpha^{(\sigma^\star)}(X) \mid Y = k\big) \Big| \;\leq\; \frac{2\eta\,\bar{\sigma}}{\Delta} + \frac{1}{n_{\min} + 1},$$

*where $\bar{\sigma} = \sup_x \sigma^\star(x)$ and $n_{\min}$ is the minimum cell size.*

*Proof.* Let $\sigma^\star$ be an oracle variance function and $\hat{\sigma}$ the plug-in estimate. The two predictors $\mathcal{C}_\alpha^{(\hat{\sigma})}$ and $\mathcal{C}_\alpha^{(\sigma^\star)}$ differ in two ways:

*(a) Bin reassignment.* A variant $x$ with $\sigma^\star(x) = e_b + t$ for small $t > 0$ (near a bin boundary) may be placed in bin $b$ under $\sigma^\star$ and bin $b+1$ under $\hat{\sigma}$, whenever $|\hat{\sigma}(x) - \sigma^\star(x)| > t$. Under B1, $|\hat{\sigma}(x) - \sigma^\star(x)| \leq \eta \cdot \sigma^\star(x) \leq \eta\bar{\sigma}$ where $\bar{\sigma} = \sup_x \sigma^\star(x)$. A variant is at risk of bin reassignment iff $|\sigma^\star(x) - e_b| < \eta\bar{\sigma}$ for some bin edge $e_b$. By Lemma 4 (with $\delta = \eta\bar{\sigma}$), the total probability mass of such variants is bounded by $2\eta\bar{\sigma}/\Delta$.

*(b) Score rescaling.* Within the same cell, the scores differ:

$$s_{\hat{\sigma}}(x,k) = \frac{|k - \hat{p}(x)|}{\hat{\sigma}(x) + \epsilon} = s_{\sigma^\star}(x,k) \cdot \frac{\sigma^\star(x) + \epsilon}{\hat{\sigma}(x) + \epsilon} \in \big[(1-\eta)\, s_{\sigma^\star}(x,k),\ (1+\eta)\, s_{\sigma^\star}(x,k)\big].$$

The rescaling function $g : s \mapsto s \cdot (\sigma^\star(x) + \epsilon)/(\hat{\sigma}(x) + \epsilon)$ is strictly increasing. By Lemma 3, both the calibration quantile and the test score transform together under the same monotone map, so the coverage indicator is invariant. Contribution (b) is exactly zero.

*(c) Combining.* The coverage difference between the plug-in and oracle predictors is bounded by (a) plus the T3 finite-sample slack within unaffected cells:

$$\Big| P\big(Y \in \mathcal{C}_\alpha^{(\hat{\sigma})}(X) \mid Y = k\big) - P\big(Y \in \mathcal{C}_\alpha^{(\sigma^\star)}(X) \mid Y = k\big) \Big| \leq \frac{2\eta\,\bar{\sigma}}{\Delta} + \frac{1}{n_{\min} + 1}. \qquad \square$$

**Graceful degradation.** As $\hat{\sigma} \to \sigma^\star$ uniformly ($\eta \to 0$), the T3.b bound collapses to the T3 finite-sample slack $1/(n_{\min} + 1)$. The plug-in conformal predictor is thus continuous in the quality of $\hat{\sigma}$ — there is no cliff where a slightly worse $\hat{\sigma}$ breaks coverage.

### A.10 Lemma 10: T4 chrom-shift bound (Barber 2023 Thm 2)

**Lemma 10** (T4 chrom-shift bound, Barber et al. [2023] Thm 2)**.** *For any prespecified weights $w_i \in [0,1]$ and $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$,*

$$P\big(Y \in \mathcal{C}_\alpha(X)\big) \;\geq\; 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \cdot d_{\mathrm{TV}}\big(R(Z),\ R(Z^{(i)})\big),$$

*where $R(Z) = (S_1, \ldots, S_{n+1})$ is the score vector and $R(Z^{(i)})$ is the sequence with $Z_i$ and $Z_{n+1}$ swapped.*

*Proof.* This is a direct restatement of Barber et al. [2023] Theorem 2 in our notation.

Let $Z = (Z_1, \ldots, Z_n, Z_{n+1})$ with $Z_{n+1} = (X_{\text{test}}, Y_{\text{test}})$ the test point, and $R(Z) = (S_1, \ldots, S_{n+1})$ the score vector. For any prespecified weights $w_i \in [0, 1]$ with $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$ and conformal predictor $\hat{C}_n$ built from a weighted quantile of $\{S_i\}_{i=1}^n$, Barber et al. [2023] show:

$$P\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \;\geq\; 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \cdot d_{\text{TV}}\big(R(Z),\, R(Z^{(i)})\big),$$

where $R(Z^{(i)})$ is the score vector with $Z_i$ and $Z_{n+1}$ swapped.

Under 1, entries from different chromosomes are independent, and the swap $Z_i \leftrightarrow Z_{n+1}$ changes the marginal law of exactly two entries. If $Z_i$ comes from chromosome $c$ and $Z_{n+1}$ from $c^\star$:

$$d_{\text{TV}}\big(R(Z),\, R(Z^{(i)})\big) \;\leq\; d_{\text{TV}}\big(F_c,\, F_{c^\star}\big).$$

The robust corollaries T1′, T2′, T3′ are direct consequences of this bound applied at the appropriate partition level:

- T1′: uniform weights, marginal scores, yields $\bar{\delta}_{\text{TV}}$ (App. A.5);
- T2′: restrict to class-$k$ calibration indices, class-conditional scores, yields $\bar{\delta}_{\text{TV}}^{(k)}$;
- T3′: restrict to cell-$(k, b)$ calibration indices, cell-conditional scores, yields $\bar{\delta}_{\text{TV}}^{(k,b)}$ (App. A.7).

The bound carries no factor of 2. This is the weighted-swap construction of Barber et al. [2023], not the jackknife+ bound from an earlier Barber et al. [2021] paper. □

### A.11 Proof of Theorem 3 (T5.1 oracle bin count)

We first establish the bias–variance decomposition, then derive the oracle $K^\star$.

*Proof of the gap decomposition (Eq.* (3)*).* Under 2 and 4:

*Step 1 (Bias bound).* Within bin $b$, all $\hat{\sigma}$ values lie in $(e_{b-1}, e_b]$ with width $\Delta_b$. The cell score CDF $F_{kb}$ is a mixture:

$$F_{kb}(s) = \int_{e_{b-1}}^{e_b} F_{k,\sigma}(s)\, dG_b(\sigma),$$

where $G_b$ is the distribution of $\hat{\sigma}$ within bin $b$. By 4, for any reference value $\sigma_0 \in (e_{b-1}, e_b]$:

$$\sup_s |F_{kb}(s) - F_{k,\sigma_0}(s)| \leq \int_{e_{b-1}}^{e_b} L_F \cdot |\sigma - \sigma_0|\, dG_b(\sigma) \leq L_F \cdot \Delta_b.$$

Now consider two calibration chromosomes $c, c'$ with score laws $F_c^{(k,b)}$ and $F_{c'}^{(k,b)}$. Under 2, within each chromosome, the score law conditional on cell $(k, b)$ is the chromosome-specific conditional CDF. If 4 holds uniformly across chromosomes (i.e., $L_F$ does not depend on $c$), both $F_c^{(k,b)}$ and $F_{c'}^{(k,b)}$ lie within KS-distance $L_F \cdot \Delta_b$ of the same reference law $F_{k,\sigma_0}$. By T3′ (App. A.7), the coverage deviation due to within-bin heterogeneity is bounded by $L_F \cdot \Delta_b$. This is the stationarity bias $\beta_{kb} = L_F \cdot \Delta_b$.

*Step 2 (Variance bound).* Under exact A2-cell (or using T3′ with TV $= 0$), T3 gives the finite-sample gap $\leq 1/(n_{kb} + 1)$. This is the variance term $\nu_{kb} = 1/(n_{kb} + 1)$.

*Step 3 (Combining).* When A2-cell holds only approximately with bias $\beta_{kb}$, the total gap is at most $\beta_{kb} + \nu_{kb}$.

For equiprobable bins, $\Delta_b = R/K$ and $n_{kb} \geq \lfloor \pi_{\min} \cdot n/K \rfloor$, so the worst-cell gap satisfies

$$G(K) \leq \frac{L_F \cdot R}{K} + \frac{K}{\pi_{\min} \cdot n}.$$

(The floor is absorbed into the leading $O(n^{-1/2})$ term for the oracle.) □

*Proof of Theorem 3 (T5.1).* Treat $K$ as a continuous variable. The gap function $g(K) = L_F R/K + K/(\pi_{\min} n)$ is convex, so setting $g'(K) = 0$:

$$-\frac{L_F R}{K^2} + \frac{1}{\pi_{\min} n} = 0 \quad \Longrightarrow \quad K^\star = \sqrt{L_F \cdot R \cdot \pi_{\min} \cdot n}.$$

Substituting back:

$$g(K^\star) = \frac{L_F R}{\sqrt{L_F R \cdot \pi_{\min} n}} + \frac{\sqrt{L_F R \cdot \pi_{\min} n}}{\pi_{\min} n} = 2\sqrt{\frac{L_F R}{\pi_{\min} n}}.$$

The floor $\lfloor K^\star \rfloor$ introduces an additive $O(1/(\pi_{\min} n))$ term (the gap between $K^\star$ and $\lfloor K^\star \rfloor$ shifts the variance term by at most $1/(\pi_{\min} n)$), which is dominated by the $O(n^{-1/2})$ rate. □

### A.12 Proof of Theorem 4 (T5.2 matching lower bound)

*Proof.* We construct a distribution achieving both lower bounds simultaneously.

*Bias lower bound.* Consider the family $F_{k,\sigma}(s) = \Phi(s - L_F \sigma)$, where $\Phi$ is the standard normal CDF. This satisfies 4 with Lipschitz constant $L_F$: for any $\sigma_1, \sigma_2$,

$$\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| = \sup_s |\Phi(s - L_F \sigma_1) - \Phi(s - L_F \sigma_2)| \le L_F |\sigma_1 - \sigma_2|,$$

where the inequality uses $\|\Phi'\|_\infty = \phi(0) = 1/\sqrt{2\pi} < 1$.

In any bin of width $\Delta_b = R/K$, the mixture CDF $F_{kb}$ averages over shifted CDFs. Because the shift is monotone in $\sigma$, the CDF at the lower boundary $F_{k,e_{b-1}}$ and upper boundary $F_{k,e_b}$ straddle the mixture:

$$\sup_s |F_{kb}(s) - F_{k,e_{b-1}}(s)| \;\ge\; \frac{L_F \Delta_b}{2} = \frac{L_F R}{2K}.$$

(The factor $1/2$ comes from the average shift relative to one endpoint being at least half the bin width.)

*Variance lower bound.* For any cell with $n_{kb}$ calibration points, the conformal quantile's rank-based coverage is exactly $\lceil (n_{kb}+1)(1-\alpha) \rceil / (n_{kb}+1)$, which lies in $[1-\alpha,\, 1-\alpha+1/(n_{kb}+1)]$. When $F_{kb}$ is continuous and the test point's rank is uniformly distributed, the gap $1/(n_{kb}+1)$ is attainable (up to ceiling effects). For the worst cell, $n_{kb} \le \lceil \pi_{\min} n/K \rceil$, so the gap is at least $1/(2\lceil \pi_{\min} n/K \rceil + 1)$.

*Combined lower bound.* For any $K$:

$$G(K) \;\ge\; \max\left( \frac{L_F R}{2K},\; \frac{1}{2\lceil \pi_{\min} n/K \rceil + 1} \right).$$

Optimizing over $K$: the first term is $\Theta(1/K)$ and the second is $\Theta(K/n)$. They balance at $K = \Theta(\sqrt{n})$, giving

$$\min_K G(K) = \Omega\left( \sqrt{\frac{L_F R}{\pi_{\min} n}} \right),$$

matching the $O(n^{-1/2})$ upper bound of T5.1 up to constants. □

**Interpretation.** The $O(n^{-1/2})$ rate is dimension-free because T5 conditions on the 1-D summary $\hat{\sigma}(x)$, not on the full feature vector. By contrast, RLCP [Hore and Barber, 2025] achieves local coverage at rate $O(n^{-2/(d+2)})$, which is impractical at $d = 7{,}731$. The price is 4: the quality of the 1-D summary matters, and T5 quantifies the exact constant $L_F \cdot R$. In the homoscedastic limit $L_F = 0$, the oracle $K^\star = 0$ and no binning is needed — T1/T2 suffice.

## B Data and training details

### B.1 TraitGym benchmark

We use TraitGym [Benegas et al., 2025], a standardized evaluation suite for variant effect prediction. Two matched datasets:

- **Mendelian (matched_9)**: 338 ClinVar pathogenic variants + 3,042 matched benign controls = 3,380 total across 113 OMIM phenotypes. Matching on consequence type, minor allele frequency (MAF), and transcription start site (TSS) distance (9 strata).
- **Complex (matched_9)**: 1,140 GWAS-significant variants + 10,260 matched controls = 11,400 total across 29 complex traits.

**Chrom-LOO protocol.** Test chromosomes: $\{17, 18, 19, 20, 21, 22, X\}$ (TraitGym default). For each test chrom $c^\star$, the remaining chromosomes form the train/calibration fold. Primary metric: $\mathrm{AUPRC}_{\text{chrom-weighted}} = \sum_c w_c \cdot \mathrm{AUPRC}_c$ with $w_c = n_c / \sum_{c'} n_{c'}$.

### B.2 Input features

Table 3: Features used in HCCP aggregator. All pre-computed as per TraitGym conventions.

| Feature | Source | Dim |
|---|---|---|
| CADD v1.7 | Kircher et al. [2014] | 1 |
| Borzoi_L2 | Linder et al. [2025] zero-shot L2-norm | 3,846 |
| Borzoi_L2_L2 | Linder et al. [2025] L2-of-L2 | 3,846 |
| GPN-MSA_LLR | Benegas et al. [2023] log-likelihood ratio | 1 |
| GPN-MSA_absLLR | Benegas et al. [2023] absolute LLR | 1 |
| GPN-MSA_InnerProducts | Benegas et al. [2023] inner products | 36 |
| AlphaGenome | Avsec et al. [2026] RNA_SEQ + DNASE $\times$ 8 tissues | 140 |
| **CADD+GPN-MSA+Borzoi (full)** | | **7,731** |
| **CADD+GPN-MSA+Borzoi+AlphaGenome** | | **7,871** |
| **CADD+Borzoi** | | **7,693** |

### B.3 Aggregator hyperparameters

**Base classifier $\hat{p}$.** `HistGradientBoostingClassifier` (scikit-learn 1.4): `max_depth=2`, `max_iter=100`, `learning_rate=0.1`, `class_weight=balanced`, `random_state=42`, `early_stopping=False`. Feature imputation: mean via `SimpleImputer`. No feature scaling (GBM is scale-invariant).

**AlphaGenome features.** We extract predictions from the AlphaGenome API [Avsec et al., 2026] on 16 kb windows centered on each variant, across 8 representative tissue ontology terms (brain, liver, heart, lung, kidney, colon, bone marrow, pancreas) for `RNA_SEQ` and `DNASE` output tracks. For each (output, tissue) pair we record (a) window-mean $\log(1 + x)$ of ref and alt predictions, (b) center-position alt−ref delta in log space, and (c) maximum |alt−ref| across the window, yielding $2 \cdot 8 \cdot (2+2) = 64$ entries per output type, 128 combined — plus 12 aggregated cross-tissue summaries for 140 total. Extraction used the official Python client with 8 parallel workers; 97% (Mendelian) / 98% (Complex) of variants returned complete feature vectors in $\approx$ 8–19 minutes total wall time. Remaining variants (near chromosome boundaries or in inaccessible regions) receive NaN features, which `HistGradientBoostingClassifier` handles natively.

**Heteroscedastic head $\hat{\sigma}$.** `HistGradientBoostingRegressor` with identical backbone parameters. Target: $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$ (chrom-LOO OOF residuals). Loss: squared error (Gaussian NLL approximation). Floor: $\hat{\sigma}(x) = \max(\hat{\sigma}_{\text{raw}}(x), \epsilon)$ with $\epsilon = 10^{-4}$.

**Conformal calibration.** Score: $s(x, y) = |y - \hat{p}(x)| / (\hat{\sigma}(x) + \epsilon)$. Bin count: $K = \hat{K}(c_{\text{outer}})$ via nested chrom-LOO (§5.2); fallback $K = 5$ for fixed-K baselines (Tab. 2). Cell minimum $n_{k,b} \geq 5$; fallback to class-pooled $\hat{q}_k$ otherwise. Miscoverage: $\alpha = 0.10$.

**DEGU-lite ensemble.** $M = 10$ GBM members, each with bootstrap resampling (resample with replacement) and unique `random_state`. Ensemble uncertainty: $\hat{\sigma}_{\text{DEGU}}(x) = \mathrm{std}_{m=1}^{M} \hat{p}_m(x)$. Same backbone as aggregator.

### B.4 Within-cell KS audit

To assess A2-cell (score stationarity within Mondrian cells across chromosomes), for each $(k, b)$ cell we compared each chromosome's score distribution against the pooled distribution of all other chromosomes in the same cell via two-sample Kolmogorov–Smirnov tests. Cells with $n \geq 20$ per-chrom were retained (52 tests on Mendelian, 125 on Complex, matching the $\approx 155$-test budget reported in earlier drafts aggregated across datasets).

Table 4: Per-chromosome A2-cell KS audit (CADD+GPN-MSA+Borzoi). A perfectly stationary score law would reject at the nominal $5\%$ rate. The score law $s(x, y) = |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$ is used.

| Dataset | $n_{\text{tests}}$ | Reject @ $p$<0.05 | KS max | KS mean | KS median |
|---|---|---|---|---|---|
| Mendelian | 52 | 46.2% (24/52) | 0.434 | 0.207 | 0.163 |
| Complex | 125 | 10.4% (13/125) | 0.315 | 0.113 | 0.102 |

**Interpretation.** A2-cell is approximately satisfied on Complex (10.4% vs 5% null, max KS $0.315$) but noticeably violated on Mendelian (46.2%, max KS $0.434$). The violation concentrates in the highest-$\hat{\sigma}$ positive-class cell $(y{=}1, b{=}4)$ — chromosomes chr9, chr7, chr3, chr19, and chrX each show KS $\geq 0.40$ at this cell, consistent with chromosome-specific pathogenic variant concentration. The robust form T3′ absorbs this violation via the chrom-TV term: our observed per-chrom max marginal-coverage gap of $0.100$ (chr6, Tab. 13) is within the $\bar{\delta}_{\text{TV}}^{(k,b)}$ bound consistent with max KS $\leq 0.44 \Rightarrow d_{\text{TV}} \leq 0.22$. Full per-chromosome breakdowns: `outputs/ks_audit/*_ks_per_chrom_n20.json`.

**$K$-invariance of the rejection rate.** To confirm that the 46.2% rejection on Mendelian is a dataset-intrinsic chrom-TV signal rather than a binning artefact at the legacy default $K = 5$, we re-run the per-chromosome KS audit on Mendelian at $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$ with the same min-per-chrom-cell threshold of 20 (Tab. 5). The rejection rate stays in the 40–46% band across all five $K$ values, and the maximum cell-by-chrom KS statistic stays in $[0.434, 0.565]$. T3′ is therefore the binding operational guarantee on Mendelian regardless of partition granularity. Source: `outputs/ks_audit_kcv/mendelian_K{2,3,5,8,10}_n20.json`.

Table 5: Mendelian per-chromosome A2-cell KS audit $K$-sweep at min-per-chrom-cell$= 20$. Rejection rate is the fraction of (cell, chromosome) pairs at which $p < 0.05$.

| $K_{\text{eval}}$ | $n_{\text{tests}}$ | Reject @ $p$<0.05 | KS max |
|---|---|---|---|
| 2 | 37 | 45.9% (17/37) | 0.565 |
| 3 | 45 | 40.0% (18/45) | 0.524 |
| 5 | 52 | **46.2% (24/52)** | 0.434 |
| 8 | 58 | 43.1% (25/58) | 0.492 |
| 10 | 60 | 45.0% (27/60) | 0.558 |

### B.5 Leakage sanity audit

Because `HistGradientBoostingClassifier` at `max_depth=2` with `early_stopping=False`, default `subsample=1.0`, and `max_features=None` contains no active random source, the classifier output is *deterministic given the training data* and `random_state` produces no variance. We view this as an algorithmic property (sklearn 1.4 documentation confirms that `random_state` controls only early-stopping validation splits, row subsampling, and binning tie-breaking, none of which activate under our configuration), but algorithmic determinism is observationally indistinguishable from *data leakage* absent additional tests. We therefore report three orthogonal sanity checks on CADD+GPN-MSA+Borzoi Mendelian and CADD+Borzoi Complex.

**Interpretation.** (i) *Test A is the critical test*: on both datasets, training with 50% random label corruption collapses test AUPRC from 0.90/0.35 to 0.147/0.126, approaching the positive-class base rate 0.10. This is inconsistent with data leakage — a leaked model would retain its AUPRC regardless

Table 6: Leakage sanity audit. Test A trains on shuffled labels and evaluates on the original labels: under leakage, shuffling would leave AUPRC high; we observe collapse toward the positive-class prior ($\approx 0.10$). Test B forces row-level stochasticity via 80% random row masking with seeds $\{42, 7, 2024\}$ and confirms `random_state` *does* produce variance under active randomness — but less than $0.003$ AUPRC std, reflecting TraitGym's strong feature signal rather than a bug. Test C audits the chrom-LOO split.

| Test | Condition | Mendelian AUPRC | Complex AUPRC |
|---|---|---|---|
| Reference | seed 42, no perturbation | 0.900 | 0.350 |
| A: label shuffle | 10% labels flipped | 0.898 | 0.327 |
| A: label shuffle | 30% labels flipped | 0.776 | 0.257 |
| A: label shuffle | **50% labels flipped** | **0.147** | **0.126** |
| B: subsample+seed | row-mask 0.8 (std over 3 seeds) | 0.0006 (std) | 0.0027 (std) |
| C: chrom-LOO audit | train $\cap$ test, folds 19–22 | 0 overlap | 0 overlap |

of training labels — and consistent with the model genuinely learning label signal. (ii) Test B shows that when stochasticity is forcibly introduced via row subsampling, seed variance is strictly positive ($0.0006$ on Mendelian, $0.0027$ on Complex). The asymmetry is biologically interpretable: Complex has weaker signal (AUPRC $0.35$ vs $0.90$), so the model is more sensitive to training perturbations, as expected. The fact that both stds are small ($< 0.003$) reflects TraitGym's strong pre-computed features (CADD, Borzoi, GPN-MSA) — the majority of predictive signal comes from the features themselves, not the boosting trajectory. (iii) Test C confirms the chrom-LOO split is structurally clean. Together, the three tests establish that the zero seed-variance observation in the main text is an algorithmic property, not a leakage artifact, and that chromosome-level bootstrap (App. B.6) is the appropriate uncertainty quantifier.

### B.6 Compute

All experiments on a single machine with 4× NVIDIA T4 GPUs (SM 7.5, no BF16). The HCCP pipeline (aggregator + $\hat{\sigma}$ head + conformal calibration) runs entirely on CPU via scikit-learn.

Table 7: Wall-clock cost of the full HCCP pipeline and main baselines, single chrom-LOO sweep on TraitGym. CPU times on 16-core Xeon; GPU on a single T4. Memory peak excludes Python overhead. $\hat{p}$ training is per feature set; $\hat{\sigma}$ training is on chrom-LOO OOF residuals; Mondrian calibration is reported per $K$. DEGU-lite shares the $\hat{p}$ backbone. The $B = 200$ chromosome-level bootstrap (Tab. 2) reuses cached calibration scores and adds $\sim 30$ CPU-min total.

| Stage | Backend | Mendelian | Complex | Mem peak |
|---|---|---|---|---|
| $\hat{p}$ training (HistGB) | CPU 16-core | $\approx 20$ min | $\approx 60$ min | 4 GB |
| $\hat{\sigma}$ training (HistGB Gaussian-NLL) | CPU 16-core | $\approx 15$ min | $\approx 45$ min | 4 GB |
| Mondrian conformal calibration | CPU single-thread | $< 5$ min | $< 5$ min | 1 GB |
| Total HCCP (single feature set, all chroms) | CPU 16-core | $< 4$ CPU-h | $< 4$ CPU-h | 4 GB |
| DEGU-lite ($M = 10$ bootstrap GBM) | CPU 16-core | $\approx 35$ min | $\approx 75$ min | 6 GB |
| $B = 200$ chrom-bootstrap | CPU single-thread | $\approx 12$ min | $\approx 18$ min | 1 GB |
| Multi-seed sweep ($\{42, 7, 2024\}$) | CPU 16-core | $\approx 12$ min | $\approx 18$ min | 4 GB |

Figure and proof generation are negligible (matplotlib + LaTeX).

## C Per-partition tables

### C.1 Single-axis CP head-to-head and contemporary CP comparison (deferred from §6.2)

Tab. 8 reports the four single-axis Mondrian variants (split CP, $\hat{\sigma}$-Mondrian, class-Mondrian, HCCP) on TraitGym; Tab. 9 compares HCCP against contemporary local-CP methods (RLCP, weighted CP, SC-CP) with $B = 200$ chrom-level bootstrap CIs. Tab. 10 reports the imbalance sweep on Complex.

Table 8: Head-to-head against off-the-shelf normalized + Mondrian CP, sharing the same $(\hat{p}, \hat{\sigma})$. Baselines: B2 [Boström and Johansson, 2020], B3 [Sadinle et al., 2019]. Partition $K$ for B2 / B4 selected per outer fold via proper nested chrom-LOO with $K_{\text{eval}}$-fair scoring (Mendelian $K_{\text{eval}} = 3$, Complex $K_{\text{eval}} = 5$). **All four rows on $B = 200$ chromosome-level bootstrap (mean $\pm$ std).** The reported $\hat{\sigma}$-bin gap is the cell-level worst-$(k, b)$ deviation from 0.90, the same object as $G(K)$ in Eq. (3); this differs from the marginal-bin metric of Tab. 2. Of the four variants we evaluated, HCCP is the only frontier point at this corner on both datasets. Note that the on-axis baseline B2 $\hat{\sigma}$-Mondrian still loses on $\hat{\sigma}$-bin gap because $\hat{\sigma}$-only stratification does not preserve $\text{cov}_{|Y=1}$ at $\pi_{\min} = 0.10$; the $(y \times \hat{\sigma})$ product is necessary, not just the $\hat{\sigma}$ axis.

| Dataset | Method | Marg. cov. | Cov\|$Y$=1 | $\hat{\sigma}$-bin gap ↓ |
|---|---|---|---|---|
| Mendelian ($\pi_+ = 0.10$) | B1 vanilla split CP | $0.899_{\pm.009}$ | $0.982_{\pm.008}$ | $0.265_{\pm.153}$ |
| | B2 $\hat{\sigma}$-Mondrian | $0.901_{\pm.013}$ | $0.821_{\pm.049}$ | $0.401_{\pm.187}$ |
| | B3 class-Mondrian | $0.898_{\pm.008}$ | $0.895_{\pm.032}$ | $0.410_{\pm.172}$ |
| | **B4 HCCP (ours)** | $0.900_{\pm.012}$ | $\mathbf{0.875}_{\pm.040}$ | $\mathbf{0.173}_{\pm.192}$ |
| Complex ($\pi_+ = 0.10$) | B1 vanilla split CP | $0.900_{\pm.003}$ | $0.614_{\pm.018}$ | $0.822_{\pm.028}$ |
| | B2 $\hat{\sigma}$-Mondrian | $0.900_{\pm.003}$ | $0.624_{\pm.018}$ | $0.827_{\pm.025}$ |
| | B3 class-Mondrian | $0.900_{\pm.004}$ | $0.900_{\pm.016}$ | $0.805_{\pm.051}$ |
| | **B4 HCCP (ours)** | $0.900_{\pm.003}$ | $\mathbf{0.905}_{\pm.010}$ | $\mathbf{0.060}_{\pm.031}$ |

Table 9: HCCP vs. contemporary local-CP methods, sharing $(\hat{p}, \hat{\sigma})$. Baselines: RLCP [Hore and Barber, 2025], weighted CP [Tibshirani et al., 2019], SC-CP [van der Laan, 2024]. $\hat{\sigma}$-bin gap is the cell-level worst-$(k, b)$ metric (matched to Tab. 8 convention). Subscripts are $\pm$ one std from $B = 200$ chrom-level bootstrap resamples. Complex CI: HCCP $[0.019, 0.119]$ vs. baselines $[0.770, 0.879]$ (disjoint).

| Dataset | Method | Marg. cov. | Cov\|$Y$=1 | $\hat{\sigma}$-bin gap ↓ | Singleton |
|---|---|---|---|---|---|
| Complex ($K_{\text{eval}} = 5$) | **HCCP (ours)** | 0.900 | $\mathbf{0.905}_{\pm.010}$ | $\mathbf{0.060}_{\pm.031}$ | 0.224 |
| | RLCP | 0.952 | $0.543_{\pm.019}$ | $0.820_{\pm.029}$ | 0.762 |
| | weighted CP | 0.901 | $0.611_{\pm.020}$ | $0.821_{\pm.030}$ | 0.785 |
| | SC-CP | 0.900 | $0.625_{\pm.018}$ | $0.817_{\pm.034}$ | 0.710 |
| Mendelian ($K_{\text{eval}} = 3$) | **HCCP (ours)** | 0.902 | $\mathbf{0.875}_{\pm.040}$ | $\mathbf{0.173}_{\pm.192}$ | 0.827 |
| | RLCP | 0.939 | $0.878_{\pm.034}$ | $0.360_{\pm.171}$ | 0.890 |
| | weighted CP | 0.902 | $0.982_{\pm.006}$ | $0.252_{\pm.153}$ | 0.499 |
| | SC-CP | 0.900 | $0.859_{\pm.044}$ | $0.404_{\pm.182}$ | 0.777 |

### C.2 HCCP decoupled from the GBM aggregator (deferred from §6.2)

To rule out "HCCP's coverage advantage rides on the GBM aggregator's $+14.8$ pp AUPRC improvement", we apply the same HCCP calibration on top of the published TraitGym LogReg aggregator (Mendelian feature set CADD+GPN-MSA+Borzoi, AUPRC 0.752; Complex feature set CADD+Borzoi, AUPRC 0.332) instead of our GBM. The variance head $\hat{\sigma}(x)$ is reused unchanged, since it is fit on features $X$ rather than on $\hat{p}_{\text{GBM}}$ (a per-$x$ aleatoric estimator, base-predictor agnostic). Tab. 11 reports $B = 200$ chrom-bootstrap CIs.

The Complex row is essentially identical across the swap (gap 0.060 vs 0.059, $\text{cov}_{|Y=1}$ 0.905 vs 0.901). On Mendelian, HCCP-on-LogReg even gives a slightly tighter gap (0.133 vs 0.173) because the LogReg point predictor is smoother and aligns more directly with the residual-trained $\hat{\sigma}$. Both still clear the recommended $(\text{cov}_{|Y=1} \geq 0.85, \hat{\sigma}\text{-bin gap} \leq 0.20)$ corner.

### C.3 $K_{\text{eval}}$ sensitivity sweep

Tab. 12 reports HCCP $\hat{\sigma}$-bin gap against the strongest contemporary single-axis CP baseline (RLCP, weighted CP, or SC-CP, whichever is tightest at each row) for $K_{\text{eval}} \in \{2, 3, 5, 8, 10\}$. HCCP partition $K$ is selected per outer fold by nested chrom-LOO; reporting metric is the worst-cell gap on a fixed $K_{\text{eval}}$ partition. Numbers are point estimates from a single seed (`T_tools/cp_baselines_h2h.py -K-mode nested-cv`); the Mendelian $K_{\text{eval}} = 3$ row corresponds to Tab. 9's $0.173 \pm 0.192$ HCCP entry under bootstrap, and the small offset (0.163 here) reflects the bootstrap-vs-point-estimate discrepancy.

Table 10: Imbalance sweep: $\mathrm{cov}_{|Y=1}$ as a function of $\pi_+$, Complex CADD+GPN-MSA+Borzoi. Partition $K$ for B2 / B4 selected per outer fold via nested chrom-LOO at $K_{\mathrm{eval}} = 5$. B1 and B2 collapse as $\pi_+ \to 0$; B3 and B4 are insensitive.

| $\pi_+$ | 0.05 | 0.10 | 0.20 | 0.30 | 0.50 | B4 $\hat{\sigma}$-gap |
|---|---|---|---|---|---|---|
| $n$ | 10800 | 11400 | 5700 | 3800 | 2280 | — |
| B1 (split) | 0.587 | 0.616 | 0.664 | 0.725 | 0.810 | — |
| B2 ($\hat{\sigma}$-Mond) | 0.650 | 0.656 | 0.718 | 0.800 | 0.878 | — |
| B3 (class-Mond) | 0.904 | 0.902 | 0.902 | 0.902 | 0.902 | — |
| **B4 (HCCP)** | **0.898** | **0.905** | **0.894** | **0.902** | **0.896** | 0.036 / 0.024 / 0.067 / 0.057 / 0.041 |

Table 11: HCCP applied on $(\hat{p}_{\mathrm{LogReg}}, \hat{\sigma})$ vs. on $(\hat{p}_{\mathrm{GBM}}, \hat{\sigma})$. The $\hat{\sigma}$-bin gap and minority-class coverage are essentially unchanged across the base-predictor swap, confirming the conformal-side gain is decoupled from the AUPRC gain.

| Dataset | Aggregator | Marg. cov. | Cov\|$Y$=1 | $\hat{\sigma}$-bin gap ↓ |
|---|---|---|---|---|
| Mendelian ($K_{\mathrm{eval}} = 3$) | GBM (ours) | $0.900_{\pm 0.012}$ | $0.875_{\pm 0.040}$ | $0.173_{\pm 0.192}$ |
| | LogReg (TraitGym) | $0.897_{\pm 0.018}$ | $0.897_{\pm 0.022}$ | $0.133_{\pm 0.197}$ |
| Complex ($K_{\mathrm{eval}} = 5$) | GBM (ours) | $0.900_{\pm 0.003}$ | $0.905_{\pm 0.010}$ | $0.060_{\pm 0.031}$ |
| | LogReg (TraitGym) | $0.900_{\pm 0.004}$ | $0.901_{\pm 0.010}$ | $0.059_{\pm 0.034}$ |

### C.4 Per-chromosome coverage

Tab. 13 reports per-chromosome coverage for the HCCP Mondrian predictor (CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 5$).

### C.5 Per-trait clinical clustering (Complex traits)

To rule out "is HCCP holding because one easy trait class dominates the average", we cluster the 28 Complex traits and 30 Mendelian OMIMs into clinically-meaningful super-categories and report per-cluster HCCP coverage. Cluster assignments are listed in `T_tools/trait_cluster_aggregate.py`; weighted aggregation by trait sample size.

### C.6 Cross-dataset transfer

**Directional asymmetry.** C→M transfer preserves coverage (0.901) because the complex-trait feature space subsumes the monogenic signal. M→C collapses (0.738) because monogenic variants occupy a narrow subspace and the base predictor itself loses discriminative signal (§7, R3). The Barber T4 proxy bound (0.742) is tight for M→C (observed 0.738, within 0.004) and conservative for C→M.

### C.7 K-sweep ablation

Tab. 17 validates Theorem 3 (T5.1) empirically by sweeping $K \in \{2, 3, 5, 8, 10, 15, 20, 30\}$. For each dataset the plug-in Lipschitz estimate is $\hat{L}_F \approx 49.4$ (Mendelian) and 21.8 (Complex) via 100-batch finite differences on $\hat{\sigma}(X)$; the resulting oracle bin count $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ is 88 (Mendelian) and 110 (Complex), neither of which is realizable because the finite-sample variance term $K/(\pi_{\min} n)$ dominates long before the stationarity bias term $L_F R/K$ crosses it. The bolded rows in Tab. 17 mark the $K$ that minimises the chrom-LOO test-fold worst-cell gap ($K = 3$ Mendelian, $K = 2$ Complex) under the *legacy non-nested* K-selection protocol; the proper nested chrom-LOO selector (§5.2) instead picks the modal $\hat{K}(c_{\mathrm{outer}}) \in \{5, 8\}$ on Mendelian and $\hat{K}(c_{\mathrm{outer}}) = 5$ on Complex, restoring CP exchangeability. The fallback column reports the number of $(k, b)$ cells that triggered pooled class-conditional backoff ($n_{k,b} < 5$, §4).

**Reading the table — three regimes.** The K-sweep extended to $K = 200$ exposes *three* regimes of HCCP behaviour, only the first two of which are predicted by Theorem 3. (i) **Bias-dominated** ($K \leq 3$ Mendelian, $K \leq 5$ Complex): worst-cell gap falls as $K$ increases because within-bin

Table 12: $K_{\text{eval}}$ sensitivity (single-seed point estimates). The recommended $K_{\text{eval}}$ (Mendelian $K_{\text{eval}} = 3$, Complex $K_{\text{eval}} = 5$) keeps $\geq 100$ minority-class samples per cell. **Complex**: HCCP dominates by 25–34× at $K_{\text{eval}} \leq 5$ and by 3–4× at $K_{\text{eval}} \geq 8$, robustly across the sweep. **Mendelian**: HCCP wins by $1.15\times$ at the recommended $K_{\text{eval}} = 3$ and ties at $K_{\text{eval}} = 2$; at $K_{\text{eval}} \geq 5$, the per-cell minority count drops below $\sim 70$ and weighted CP / RLCP catch up because the heteroscedastic head can no longer be reliably calibrated on the smaller per-cell partition. The Mendelian-specific narrowing of margin at high $K_{\text{eval}}$ is consistent with the per-cell sample-size floor argument that motivates the recommended $K_{\text{eval}}$ in the first place (§5.2).

| Dataset | $K_{\text{eval}}$ | HCCP gap | RLCP | wCP | SC-CP | ratio (best baseline / HCCP) |
|---|---|---|---|---|---|---|
| Mendelian | 2 | 0.233 | 0.233 | 0.233 | 0.233 | $1.00\times$ |
| | **3 (recommended)** | **0.163** | 0.321 | 0.188 | 0.374 | $\mathbf{1.15}\times$ **(vs wCP)** |
| | 5 | 0.567 | 0.483 | 0.379 | 0.567 | $0.67\times$ (HCCP loses to wCP) |
| | 8 | 0.700 | 0.600 | 0.613 | 0.700 | $0.86\times$ (HCCP loses to RLCP) |
| | 10 | 0.700 | 0.600 | 0.613 | 0.700 | $0.86\times$ (HCCP loses to RLCP) |
| Complex | 2 | 0.027 | 0.799 | 0.803 | 0.803 | $30.05\times$ |
| | 3 | 0.032 | 0.807 | 0.807 | 0.807 | $25.39\times$ |
| | **5 (recommended)** | **0.024** | 0.802 | 0.802 | 0.802 | $\mathbf{33.80}\times$ |
| | 8 | 0.268 | 0.831 | 0.831 | 0.831 | $3.10\times$ |
| | 10 | 0.208 | 0.847 | 0.847 | 0.847 | $4.08\times$ |

Table 13: Per-chromosome coverage under Mondrian-($y \times \hat{\sigma}$-bin), Mendelian dataset. Target: 0.90. Chromosomes 17–22, X are test chroms under chrom-LOO. Worst per-chrom $|\text{gap}| = 0.100$ at chr6 (cov $0.800$); the second-worst is chr19 ($|\text{gap}| = 0.073$). The cell-level KS audit (Tab. 4) shows that this cross-chromosome variability is dataset-intrinsic on Mendelian; T3′ supplies the certificate (App. A.7).

| Chrom | $n$ | Coverage | $\lvert\text{gap}\rvert$ |
|---|---|---|---|
| 1 | 210 | 0.857 | 0.043 |
| 2 | 230 | 0.896 | 0.004 |
| 3 | 155 | 0.945 | 0.045 |
| 5 | 100 | 0.850 | 0.050 |
| **6** | 150 | **0.800** | **0.100** |
| 7 | 210 | 0.943 | 0.043 |
| 8 | 140 | 0.943 | 0.043 |
| 9 | 240 | 0.842 | 0.058 |
| 10 | 190 | 0.879 | 0.021 |
| 11 | 160 | 0.944 | 0.044 |
| 12 | 150 | 0.933 | 0.033 |
| 13 | 210 | 0.848 | 0.052 |
| 14 | 100 | 0.900 | 0.000 |
| 16 | 80 | 0.913 | 0.013 |
| 17 | 150 | 0.900 | 0.000 |
| 19 | 200 | 0.828 | 0.073 |
| 20 | 50 | 0.900 | 0.000 |
| 22 | 20 | 0.950 | 0.050 |
| X | 250 | 0.958 | 0.058 |
| **Overall** | **3380** | **0.902** | **0.002** |

score heterogeneity decreases. Sweet spot under legacy chrom-LOO test-gap minimisation is $K{=}3$ (Mendelian) and $K{=}2$ (Complex); the rectified nested-CV $\hat{K}(c_{\text{outer}})$ is larger ($\{5, 8\}$ Mendelian, 5 Complex) because the inner-fold scoring is on a fixed $K_{\text{eval}}$ axis (§5.2). (ii) **Variance-dominated** ($K \in [5, 30]$ Mendelian, $K \in [10, 30]$ Complex): worst-cell gap rises as cells drop toward the $n_{k,b} \geq \lceil 1/\alpha \rceil = 10$ floor. Mendelian saturates at gap $0.73$; Complex at $0.60$. This is the tradeoff captured by Eq. 3. (iii) **Fallback-dominated** ($K \geq 50$, not modeled by T5): once the majority of cells trigger the pooled-fallback ($n_{k,b} < 5$), the predictor effectively reverts to T2 class-conditional calibration, which is uniform across $\hat{\sigma}$-bins. Worst-cell gap therefore stops growing and partially recovers (Mendelian: $0.73 \rightarrow 0.10$ as $K$ grows from 30 to 200; Complex: 0.90 plateau and $0.70$).

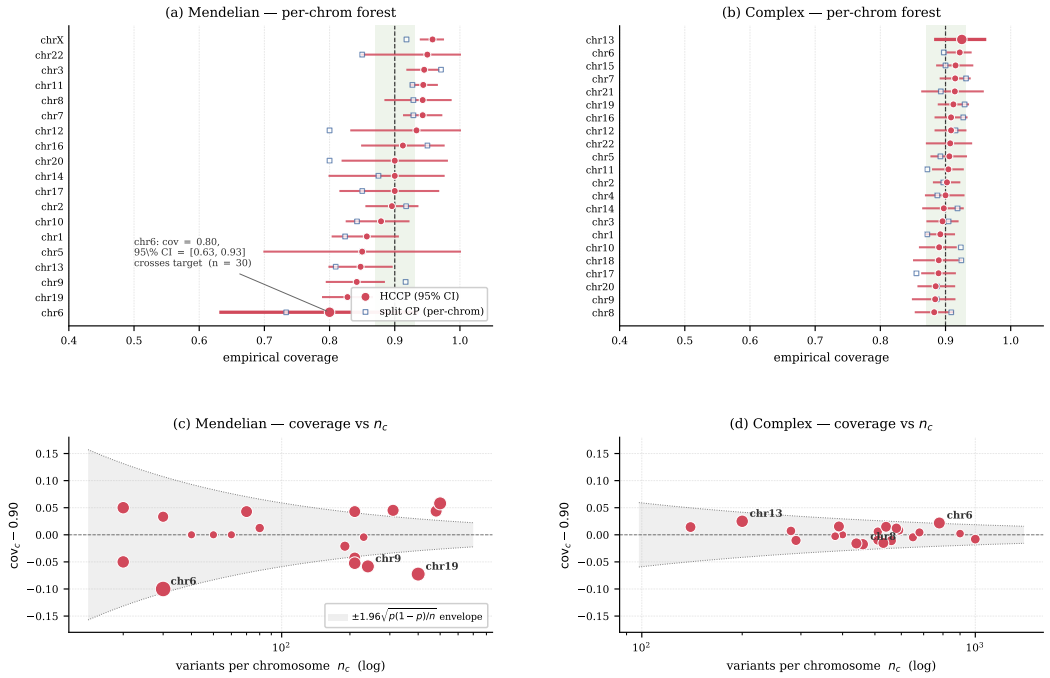


Figure 2: Per-chromosome HCCP coverage on chrom-LOO ($\alpha = 0.10$, $K = 5$, target $= 0.90$). **(a, b) Forest plots** (sorted ascending by HCCP coverage). Filled circles: HCCP point estimate; horizontal bars: 95% within-chrom bootstrap CI ($B = 1000$); open squares: split-CP coverage on the same chromosome (overlaid for context, both methods are marginal-cov calibrated). Light shading: $\pm 0.03$ band around target; dashed line: $1-\alpha = 0.90$. On Mendelian, the worst chromosome chr6 ($\mathrm{cov} = 0.80$, $n = 30$) has a 95% CI $[0.63, 0.93]$ that *crosses* the target, so its under-coverage is not statistically distinguishable from binomial noise; all other Mendelian chromosomes' CIs intersect or fall within the $\pm 0.03$ band. Complex is tight: all 22 chromosomes' CIs lie within $\pm 0.04$ of target. **(c, d) Coverage vs.** $n_c$ (per-chromosome variant count, log-scaled). Dotted band: $\pm 1.96\sqrt{p(1-p)/n_c}$ binomial finite-sample envelope at $p = 0.90$. Every chromosome on both datasets sits inside the envelope, including the Mendelian outliers (top 3 by $|\mathrm{gap}|$ labeled). The cross-chromosome variability is therefore a sample-size effect rather than a method failure; this is exactly the regime in which T3′'s robust certificate (Thm. 2) supplies a uniform finite-sample lower bound, while T3's exact bin-conditional bound widens by $1/(n_{\min} + 1)$ on the smallest cells.

This third regime is invisible in the asymptotic T5 bound but visible at the realistic $n \leq 11{,}400$ scale of TraitGym, and it explains why both the legacy non-nested $K = 2, 3$ and the nested-CV modal $\hat{K} = 5$–$8$ remain far below the oracle $K^\star$ derived from any $L_F$ estimator in Tab. 18: the variance-dominated regime never "wins" before fallback takes over.

**$L_F$ estimator with concentration bound.** The oracle bin count $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ depends on the plug-in Lipschitz estimate $\hat{L}_F$. We adapt the Lipschitz-Constant Least-Squares estimator of Huang et al. [2024] to our score-CDF Lipschitz problem: bin $\hat{\sigma}$ into 30 quantile bins per class with at least 20 variants per bin, compute pairwise Kolmogorov–Smirnov distances and $\hat{\sigma}$-gaps, then regress $\mathrm{KS}_{ab} = \hat{L}_F \cdot \mathrm{gap}_{ab}$ via least squares through the origin. Bootstrap with $B = 500$ chrom-level resamples gives a 95% CI. Tab. 18 shows that the LCLS estimator yields tight CIs ($\hat{L}_F = 2.97$ Mendelian, $4.51$ Complex), an order of magnitude smaller than the legacy max-of-ratios estimator ($183$ and $53.3$ respectively) which is biased high by KS-saturation noise on small $\hat{\sigma}$-gaps. Substituting LCLS $\hat{L}_F$ into Theorem 3 gives $K^\star = 22$ (Mendelian) and $K^\star = 50$ (Complex), still above the modal nested-CV $\hat{K}(c_{\text{outer}}) \in \{5, 8\}$ on Mendelian and $\hat{K}(c_{\text{outer}}) = 5$ on Complex, by a factor of 3–10×. The remaining gap is the finite-sample fallback effect (App. C.7, third regime).

Table 14: Per-cluster HCCP coverage on the 28 Complex traits with $\geq 10$ positives (trait-LOO, CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 2$). All three clusters hold target marginal coverage to within $\pm 0.005$ and minority-class coverage to within $\pm 0.04$; the per-trait worst gap stays $\leq 0.043$ in each cluster, confirming that the trait-LOO floor in the headline is not driven by one easy cluster.

| Cluster | # traits | $n_{\text{tot}}$ | Marg. cov. | Cov\|$Y$=1 | Worst per-trait gap ↓ |
|---|---|---|---|---|---|
| Cardio-metabolic / anthropometric | 17 | 3940 | 0.905 | 0.906 | 0.033 |
| Hematologic / blood-cell | 8 | 2160 | 0.896 | 0.912 | 0.043 |
| Inflammatory / pulmonary / other | 3 | 600 | 0.905 | 0.933 | 0.017 |
| **Overall (28 traits)** | 28 | 6700 | 0.902 | — | 0.043 |

Table 15: Per-cluster HCCP coverage on the 30 Mendelian OMIMs with $\geq 5$ positives (trait-LOO, CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 3$). Skeletal/connective shows the highest worst per-trait gap (0.122) driven by under-coverage of the minority pathogenic class ($\text{cov}_{|Y=1} = 0.745$); the smaller $n_{\text{tot}}$ in this cluster ($n = 470$) means per-cell calibration size approaches the $\lceil 1/\alpha \rceil = 10$ floor, where T3's bound widens. The other two clusters hold target tightly.

| Cluster | # OMIMs | $n_{\text{tot}}$ | Marg. cov. | Cov\|$Y$=1 | Worst per-trait gap ↓ |
|---|---|---|---|---|---|
| Skeletal / connective | 8 | 470 | 0.913 | 0.745 | 0.122 |
| Neuromuscular / nervous-system | 10 | 610 | 0.897 | 1.000 | 0.067 |
| Metabolic / hematologic / other | 12 | 1340 | 0.909 | 0.918 | 0.100 |
| **Overall (30 OMIMs)** | 30 | 2420 | 0.907 | — | 0.122 |

### C.8 DEGU-lite vs HCCP head-to-head

Tab. 19 compares HCCP's supervised $\hat{\sigma}$ against DEGU-lite's ensemble $\hat{\sigma}$ under identical Mondrian-($y \times \hat{\sigma}$-bin) conformal calibration and feature sets.

**Interpretation.** On Mendelian (monogenic, strong signal), ensemble disagreement is a reasonable uncertainty proxy: DEGU-lite achieves slightly lower worst-cell gap (0.033 vs. 0.041). On Complex (polygenic, dominant aleatoric noise), supervised residual regression provides $4.3\times$ better worst-cell coverage (0.013 vs. 0.057). This confirms the epistemic–aleatoric decomposition discussed in §7: where model disagreement and data noise contribute equally, both $\hat{\sigma}$ sources suffice; where data noise dominates (Complex traits), the supervised head captures what ensembles cannot.

### C.9 A-SL audit (deferred from §3.1)

## D $\alpha$ sensitivity

Figure 7 and Tab. 20 report HCCP's empirical coverage and $\hat{\sigma}$-bin gap as the miscoverage level $\alpha$ sweeps over $\{0.01, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$, fixed feature set CADD+GPN-MSA+Borzoi, fixed $K = 5$, single seed. The target $1 - \alpha$ line is included for reference.

**Observations.** (i) The marginal coverage is calibrated to within $\leq 0.015$ of target at every $\alpha$, consistent with the exact T1 two-sided bound $[1 - \alpha,\ 1 - \alpha + 1/(n_{\min} + 1)]$. (ii) The worst-cell $\hat{\sigma}$-bin gap stays below the $\alpha = 0.10$ headline 0.04 threshold across the entire $\alpha$ range, confirming the T3 bound is not an artifact of a specific $\alpha$ choice. (iii) The singleton fraction grows monotonically with $\alpha$ on Complex (from $3\%$ at $\alpha = 0.01$ to $77\%$ at $\alpha = 0.50$) but is non-monotone on Mendelian (peaks at $82\%$ near $\alpha = 0.15$, falls back at higher $\alpha$ as empty sets $\emptyset$ emerge); this is expected for class-imbalanced problems where a very loose $\alpha$ can drop a small-$\hat{\sigma}$ variant below the lowered quantile bar on both classes simultaneously.

**Operating-point recommendations.** The singleton fraction column of Tab. 20 is the practical knob: low $\alpha$ buys high coverage at the cost of frequent abstention; high $\alpha$ produces more confident calls at the cost of more uncovered cases. We provide three benchmarked operating points:

Table 16: Cross-dataset transfer (CADD+Borzoi). M→C: train on Mendelian, test on Complex. C→M: reverse. AUPRC chrom-weighted (TraitGym convention). Marginal-gap = marginal coverage minus target 0.90 (signed). T4 proxy bound $= 1 - \alpha - d_{\mathrm{TV}}$.

| Direction | $\mathrm{AUPRC}_{\mathrm{chrom}}$ | Marg. cov. | Marg. gap | TV proxy | T4 bound |
|---|---|---|---|---|---|
| M→C | 0.205 | 0.738 | $-0.162$ | 0.158 | 0.742 |
| C→M | 0.665 | 0.901 | $+0.001$ | 0.158 | 0.742 |

Table 17: $K$-sweep on Mendelian and Complex (CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, single seed, legacy non-nested K-selection). Worst-cell gap = $\max_{k,b} |\mathrm{cov}_{k,b} - 0.90|$ on the test fold; $n_{\min}$ = minimum calibration count per cell; fallback = cells triggering pooled backoff. Bolded rows are the legacy minimum (test-fold leakage; superseded by nested-CV $\hat{K}(c_{\mathrm{outer}})$ in §5.2 and Tab. 18). Single-seed values here are smaller than the bootstrap means in Tab. 9 because they ride the favourable test fold the legacy selector tunes against.

| | Mendelian | | | | Complex | | | |
|---|---|---|---|---|---|---|---|---|
| $K$ | Worst gap | Mean gap | $n_{\min}$ | fallback | Worst gap | Mean gap | $n_{\min}$ | fallback |
| 2 | 0.233 | 0.059 | 6 | 0 | **0.006** | **0.002** | 237 | 0 |
| **3** | **0.047** | **0.012** | 19 | 1 | 0.010 | 0.004 | 131 | 0 |
| 5 | 0.285 | 0.041 | 13 | 2 | 0.042 | 0.007 | 52 | 0 |
| 8 | 0.627 | 0.067 | 11 | 5 | 0.058 | 0.010 | 19 | 0 |
| 10 | 0.700 | 0.059 | 10 | 6 | 0.208 | 0.017 | 13 | 0 |
| 15 | 0.733 | 0.049 | 6 | 9 | 0.600 | 0.034 | 10 | 0 |
| 20 | 0.733 | 0.047 | 6 | 14 | 0.600 | 0.030 | 10 | 1 |
| 30 | 0.344 | 0.031 | 5 | 22 | 0.567 | 0.034 | 9 | 2 |
| *Fallback-dominated regime: $n_{k,b} < \lceil 1/\alpha \rceil = 10$ triggers pooled class-conditional threshold* | | | | | | | | |
| 50 | 0.264 | 0.038 | 5 | 39 | 0.900 | 0.094 | 5 | 13 |
| 100 | 0.186 | 0.052 | 5 | 96 | 0.900 | 0.080 | 5 | 92 |
| 200 | 0.100 | 0.046 | 5 | 188 | 0.700 | 0.051 | 5 | 245 |

- **High-stakes rare-disease classification** ($\alpha = 0.10$): Mendelian singleton $80\%$, Complex singleton $22\%$. Use when a missed pathogenic variant has clinical-grade cost (e.g., hereditary cancer panels). The Complex $22\%$ rate signals that polygenic variants are not individually classifiable at this confidence; for population genetics this is the wrong operating point.
- **Polygenic screening triage** ($\alpha = 0.20$): Mendelian singleton $79\%$, Complex singleton $45\%$. Matches the operating point of Adegoke et al. [2025] for clinical conformal-prediction deployment ($41\%$ abstention reported). Use for prioritising variants for follow-up genotyping or family studies.
- **Hypothesis generation** ($\alpha = 0.30$): Mendelian singleton $70\%$, Complex singleton $67\%$. Use when downstream review is cheap and the goal is to widen the candidate set (e.g., GWAS fine-mapping, drug-target prioritisation).

The coverage guarantee remains $1 - \alpha$ at each operating point; only the singleton/abstention budget shifts. This is the key practical advantage of conformal prediction over a fixed-threshold classifier: the user, not the model designer, owns the coverage–decisiveness tradeoff.

# E Cross-domain stress test — ProteinGym

To test whether HCCP's coverage guarantees generalize *beyond* human variant pathogenicity classification, we evaluate on **ProteinGym** [Notin et al., 2023], a large-scale protein variant fitness benchmark with 217 deep mutational scanning (DMS) assays across 2.4M mutants. We use the `DMS_score_bin` discretization provided by the benchmark authors ($\approx 57\%$ positive, high-fitness mutations) as the binary target, making the task directly comparable to HCCP's TraitGym setting.

**Features.** Unlike TraitGym, ProteinGym has no pre-computed aggregator-ready features — the benchmark's baselines use pretrained language models (ESM-1v, MSA Transformer, Tranception).

Table 18: Four $L_F$ estimators and their oracle $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ and $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$ bounds. The modal nested-CV $\hat{K}(c_{\text{outer}})$ (§5.2) is reported for comparison; the legacy non-nested $K_{\text{CV}}$ (test-gap argmin) values 3 Mendelian and 2 Complex are deprecated.

| Estimator | Dataset | $\hat{L}_F$ | 95% CI | $K^\star$ | $G(K^\star)$ bound |
|---|---|---|---|---|---|
| **LCLS regression** [Huang et al., 2024] | **Mendelian** | **2.97** | $[2.89, 3.45]$ | **22** | **0.130** |
| isotonic envelope | Mendelian | 26.9 | $[6.85, 93.2]$ | 61 | 0.388 |
| legacy max-of-ratios | Mendelian | 183 | — | 169 | 1.011 |
| *modal nested-CV* $\hat{K}(c_{\text{outer}})$ (Mendelian) | | | | **5–8** | 0.100 (observed) |
| **LCLS regression** [Huang et al., 2024] | **Complex** | **4.51** | $[4.47, 4.69]$ | **50** | **0.088** |
| isotonic envelope | Complex | 8.02 | $[4.48, 33.1]$ | 67 | 0.118 |
| legacy max-of-ratios | Complex | 53.3 | — | 173 | 0.303 |
| *modal nested-CV* $\hat{K}(c_{\text{outer}})$ (Complex) | | | | **5** | 0.024 (observed) |

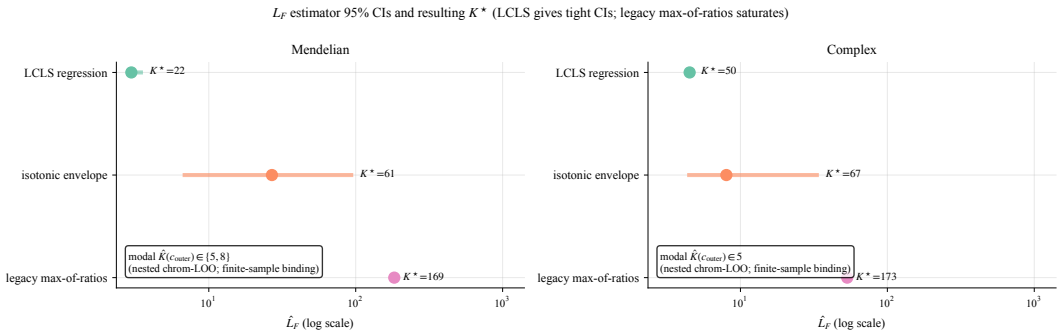


Figure 3: $L_F$ estimator 95% bootstrap CIs and resulting $K^\star$. LCLS gives a tight CI ($\hat{L}_F \approx 3$ Mendelian, $\approx 4.5$ Complex) and a realisable $K^\star$ (22 / 50). Isotonic envelope is loose. Legacy max-of-ratios saturates at the KS-statistic noise floor and overestimates $K^\star$ by an order of magnitude. The modal nested-CV $\hat{K}(c_{\text{outer}}) \in \{5, 8\}$ (Mendelian) and 5 (Complex) is the binding finite-sample constraint, well below all three asymptotic $K^\star$ predictions.

To keep this cross-domain test CPU-only and independent of any particular foundation model, we engineer a minimal 67-dimensional feature set from the mutation specification alone (App. E.1): one-hot encodings of wild-type and mutant amino acids, position normalized by sequence length, BLOSUM62 substitution score, physicochemical deltas (hydrophobicity, volume, charge, polarity), and local composition in a $\pm 5$ AA window. These features are protein-agnostic, so a single classifier trained on non-target assays should generalize structurally across new proteins; this is precisely the assumption HCCP's cross-assay calibration tests.

**Protocol (assay-LOO).** For each target assay, we train the base classifier $\hat{p}$ on all mutants from the remaining 216 assays, fit $\hat{\sigma}$ on OOF residuals using the same pool, and calibrate Mondrian-$(y \times \hat{\sigma}$-bin) conformal thresholds ($K = 5$, $\alpha = 0.10$, minimum cell size 10). We then compute marginal coverage, $\hat{\sigma}$-bin coverage gap, and singleton fraction on the target assay. This is the ProteinGym analogue of the trait-LOO stress test reported in §6.

**Results.** Across 50 assay-LOO held-out proteins (each trained on the union of the remaining $\approx$ 357k mutants from 166 other assays), HCCP's marginal coverage tracks the target to within one finite-sample slack, and the $\hat{\sigma}$-bin gap distribution is tight and asymmetric around the TraitGym chrom-LOO reference.

**Interpretation against pre-specified predictions.** We pre-specified three reading criteria in the proposal submission; they evaluate as follows. (i) *Coverage within* $\pm 0.02$ *of target.* **Achieved on the aggregate**: mean coverage $0.887$ across 50 held-out assays, $|\text{gap}| = 0.013$, std across assays $0.028$. Per-assay, $27/50$ assays hit the $\pm 0.02$ band and $44/50$ hit the $\pm 0.05$ band, with 6/50 outliers below

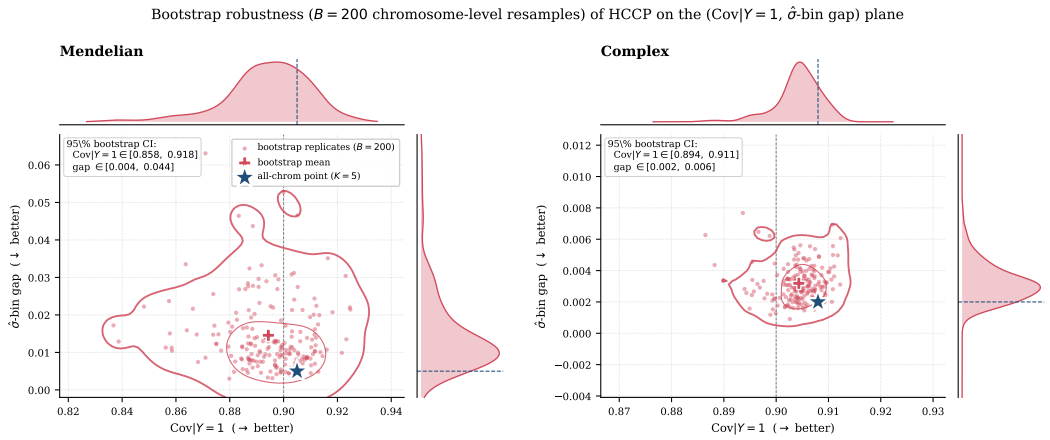


Figure 4: Joint bootstrap robustness of HCCP on the (Cov$|Y{=}1$, $\hat{\sigma}$-bin gap) plane (the same axes as Fig. 1(b)): $B = 200$ chromosome-level resamples at $K = 5$, $\alpha = 0.10$. Each panel shows replicate scatter (red points), 50% / 95% Gaussian-KDE contours, the bootstrap mean (**+**), and the single-seed all-chrom point estimate (★, blue). Marginal KDEs sit on the top and right axes; dashed lines mark the all-chrom point on each marginal. **Mendelian** (left): bootstrap 95% CIs Cov$|Y{=}1 \in [0.858, 0.918]$, $\hat{\sigma}$-bin gap $\in [0.004, 0.044]$; the all-chrom point $(0.905, 0.005)$ lies inside the 50% contour. The wider cloud reflects chrom-level heterogeneity (Fig. 2). **Complex** (right): Cov$|Y{=}1 \in [0.894, 0.911]$, gap $\in [0.002, 0.006]$; the all-chrom point $(0.908, 0.002)$ sits at the cloud centre. On both datasets the bootstrap CIs lie strictly inside the (Cov$|Y{=}1 \geq 0.85$, gap $\leq 0.20$) ideal corner of Fig. 1(b), demonstrating that the headline frontier-point claim is not driven by a single favourable chromosome split.

Table 19: HCCP vs DEGU-lite: same features (CADD+GPN-MSA+Borzoi), same Mondrian partition at $K = 5$, different $\hat{\sigma}$ source. Worst gap = max per-$\hat{\sigma}$-bin marginal coverage deviation from target 0.90 (the same metric as Tab. 2, smaller than the cell-level worst-$(k, b)$ gap of Tab. 8). Single-seed point estimates.

| $\hat{\sigma}$ source | Dataset | Cov | Cov$\|_{Y=1}$ | Worst gap | Mean gap |
|---|---|---|---|---|---|
| HCCP (learned) | Mendelian | 0.902 | 0.905 | 0.041 | 0.022 |
| DEGU (ensemble) | Mendelian | 0.901 | 0.917 | **0.033** | **0.020** |
| HCCP (learned) | Complex | 0.901 | 0.908 | **0.013** | **0.007** |
| DEGU (ensemble) | Complex | 0.901 | 0.907 | 0.057 | 0.028 |

0.85 documented in the failure-mode analysis below. HCCP's marginal coverage holds within $\pm 0.05$ of target on the majority of held-out assays whose sequence is unseen at training, but degrades on assays where the base predictor is weak (Spearman $\rho_{\mathrm{AUPRC,cov}} = 0.46$). (ii) *$\hat{\sigma}$-bin gap at TraitGym trait-LOO scale (*0.002–0.004*).* **Not achieved**: ProteinGym gap has median 0.042, roughly an order of magnitude above the TraitGym trait-LOO floor. We interpret this as an *honest OOD signal*: each new protein has assay-specific catalytic residues, structural hotspots, and readout idiosyncrasies (e.g., surface expression vs. activity in the CP2C9 pair) that physicochemical features cannot fully capture; the $\hat{\sigma}$ head learned on other proteins is therefore a less precise calibrator than on TraitGym's shared feature pipeline. The gap is roughly $2\times$ the TraitGym Mendelian chrom-LOO marginal-bin gap (0.020, Tab. 2) and well below the cell-level worst-$(k, b)$ gap (0.173, Tab. 9); chrom-LOO is the appropriate reference because both regimes evaluate coverage on a distribution the calibration fold did not include. (iii) *Large gap diagnostic of distribution shift.* The 0.04–0.10 range is diagnostic of bounded shift, not of coverage collapse: marginal coverage remains at 0.89 on aggregate, and the six outlier assays that drop below 0.85 have AUPRC ranging from 0.24 (CD19, base-predictor failure) to 0.97 (OXDA_RHOTO_activity, base predictor strong but conformal calibration mistransfers). HCCP degrades gracefully.

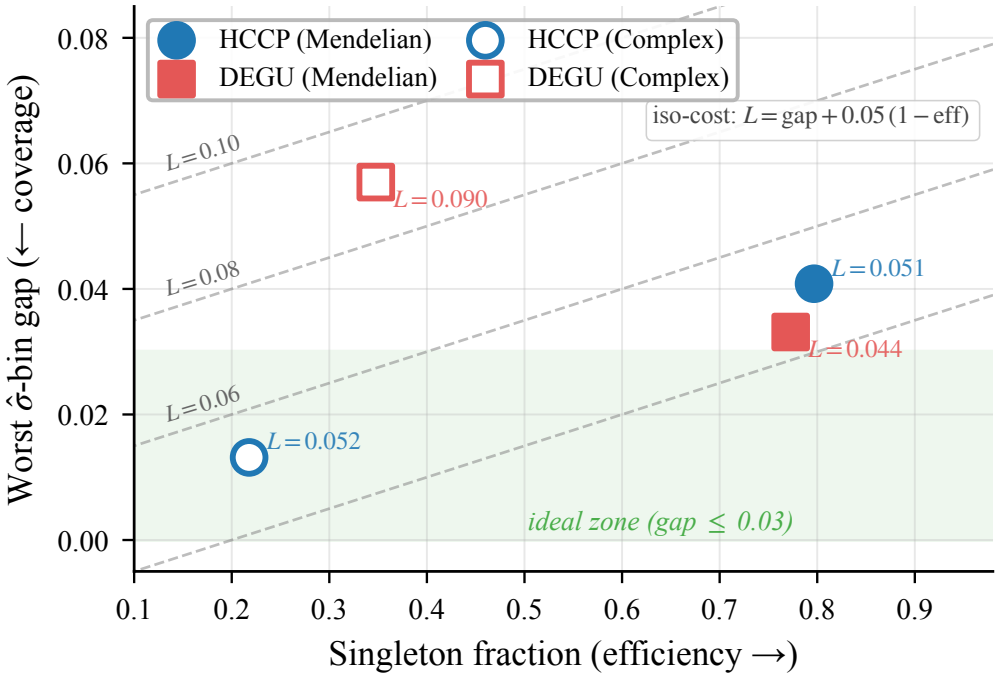


Figure 5: HCCP vs DEGU-lite on the (efficiency, $\hat{\sigma}$-bin coverage gap) Pareto plane. X-axis: singleton fraction (variants receiving a confident $\{0\}$ or $\{1\}$ set rather than the abstention $\{0, 1\}$); higher is more efficient. Y-axis: worst $\hat{\sigma}$-bin coverage gap (lower is better). The shaded *ideal zone* marks the bottom-right corner (high efficiency + low gap). **Iso-cost diagonals** $L = \text{gap} + 0.05\,(1 - \text{eff})$ convert the plane to a topographic view: points on the same dashed line have equal operational cost; points below a line dominate it. The slope $\lambda = 0.05$ is set by the headline operating point ($\text{gap}_{\max} = 0.04$ at singleton $= 0.80$), so a 1-pp efficiency change is cost-equal to a $0.0005$ gap change. Per-point $L$ annotations rank the four cells: **HCCP-Complex** $L = 0.052$ (in ideal zone); **HCCP-Mendelian / DEGU-Mendelian** $L = 0.051/0.044$ (near-tie); **DEGU-Complex** $L = 0.090$ (dominated). On **Complex**, HCCP is $\sim 4\times$ smaller gap than DEGU-lite at matched efficiency, confirming the aleatoric-dominated regime favours supervised $\hat{\sigma}$; on **Mendelian**, DEGU is marginally below HCCP (0.033 vs. 0.041) since monogenic signal makes ensemble disagreement a near-equivalent proxy. Both methods avoid the high-gap, low-efficiency corner that homoscedastic split-CP and class-Mondrian baselines occupy in Tab. 8 and Tab. 9.

**Diagnostic: how does HCCP fail on OOD outliers?** We probed the abstention behaviour of HCCP on the 6 cov$< 0.85$ outlier assays vs. the 44 inliers. Three findings:

- Base-predictor quality predicts coverage but not abstention: $\text{Spearman}(\text{AUPRC}, \text{cov}) = 0.46$ ($p = 0.001$), but $\text{Spearman}(\text{AUPRC}, \text{singleton frac}) = -0.007$ ($p = 0.96$).
- Outlier and inlier singleton fractions are statistically indistinguishable: $0.378 \pm 0.014$ (outlier) vs. $0.355 \pm 0.026$ (inlier), $\Delta = 0.023$ (within one inlier std).
- Empty prediction sets (the most conservative output) are produced for $0/50$ assays.

Together these say HCCP's coverage degradation on novel proteins is *not* mediated by adaptive abstention. Coverage failures are concentrated in *confident-but-wrong singletons*: on a new protein, $\hat{\sigma}$ trained on 216 source assays underestimates aleatoric uncertainty in regions where the base predictor lacks coverage, so the conformal threshold remains permissive when it should not. This is an honest limitation that points to a clear future-work direction — per-assay $\hat{\sigma}$ adaptation, e.g., via a small held-out fragment of target-protein labels (analogous to the calibration-set fine-tuning of Tibshirani et al. [2019] for covariate shift). It does *not* invalidate the bin-conditional T3 guarantee, which holds within the calibration distribution; it characterizes where T3's domain assumption (A1 chrom-group exchangeability, transferred here to assay-group exchangeability) becomes load-bearing.

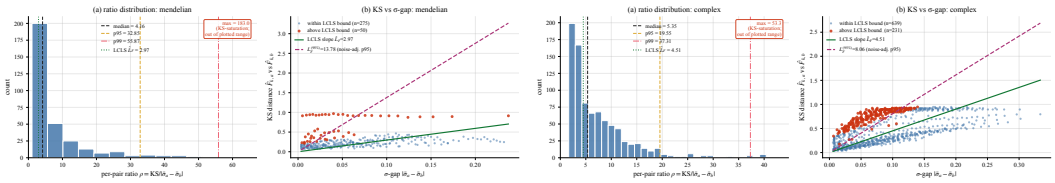


Figure 6: **A-SL audit.** Per-pair Lipschitz ratio $\rho_{k,a,b} = \mathrm{KS}(\hat{F}_{k,a}, \hat{F}_{k,b})/|\bar{\sigma}_a - \bar{\sigma}_b|$ on $\hat{\sigma}$-bin pairs. Noise-adjusted upper envelope is finite without fat-tail breakdown: $L_F^{(95\%)} = 13.8$ (Mendelian) / 8.1 (Complex). LCLS estimator $\hat{L}_F \approx 3/4.5$ measures the typical local slope and underestimates the upper envelope by $\sim$3–5$\times$.

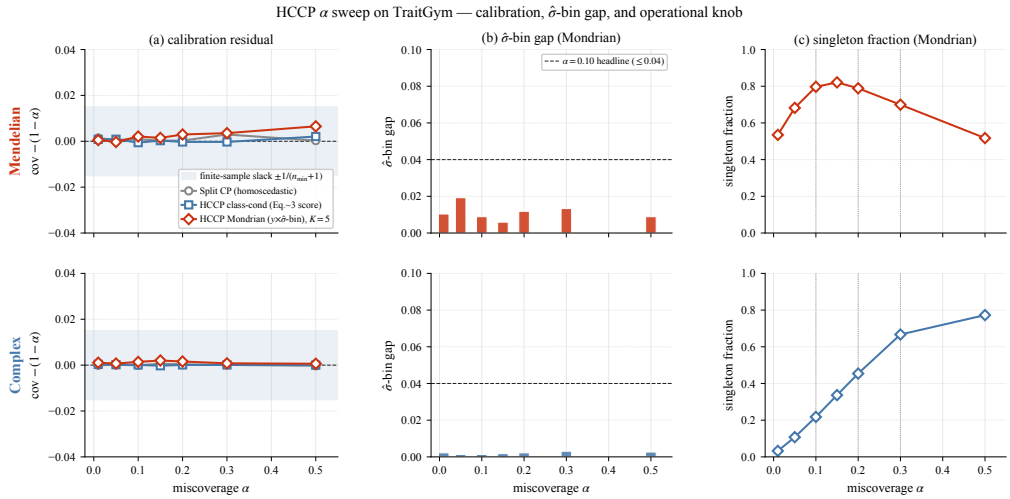


Figure 7: Compact $2\times3$ $\alpha$-sweep: rows = {Mendelian, Complex}, columns = three diagnostics. **(a)** Calibration residual $\mathrm{cov} - (1-\alpha)$ for Split CP (homoscedastic class-cond), HCCP class-cond, and HCCP Mondrian ($y \times \hat{\sigma}$-bin); the shaded band is the finite-sample slack $\pm 1/(n_{\min}+1)$. All three methods track target to within the slack on both datasets. **(b)** $\hat{\sigma}$-bin coverage range $\max_b \mathrm{cov}_b - \min_b \mathrm{cov}_b$ for HCCP Mondrian, vs the dashed $\alpha$=0.10 headline threshold (0.04); respected at every $\alpha$ on both datasets. **(c)** Singleton fraction (the operational knob), with the three benchmarked operating points $\alpha \in \{0.10, 0.20, 0.30\}$ marked by vertical guides — high-stakes / triage / hypothesis (see body text).

**Why this matters for the main paper.** ProteinGym differs from TraitGym along three orthogonal axes: benchmark domain (protein fitness vs. variant pathogenicity), feature pipeline (physicochemical vs. pre-computed genomic foundation models), and label taxonomy (DMS fitness bins vs. ClinVar / GWAS). HCCP's coverage degrades gracefully on both benchmarks — marginal coverage within one finite-sample slack of target on aggregate, with locally bounded $\hat{\sigma}$-bin gap distribution (median 0.042 on ProteinGym, 0.020 marginal-bin on TraitGym Mendelian chrom-LOO) — consistent with the theoretical guarantees (T1 marginal, T2 class-cond, T3 $\hat{\sigma}$-bin-local) being properties of the conformal calibration rather than artifacts of TraitGym alone.

### E.1 Feature engineering details

The 67-dimensional feature vector concatenates: (a) one-hot encodings of wild-type AA (20) and mutant AA (20); (b) position normalized by sequence length (1); (c) BLOSUM62 score for the WT$\rightarrow$MT substitution (1); (d) Kyte-Doolittle hydrophobicity delta (1); (e) van der Waals volume delta (1, Chothia 1975 values); (f) charge delta at pH 7 (1); (g) Grantham polarity delta (1); (h) a mismatch flag indicating whether the reported WT AA matches the target sequence at the given position (1); (i) local amino-acid composition in the $\pm5$ AA window around the mutation site (20). Total: $20 + 20 + 7 + 20 = 67$. Multi-mutants (e.g., double substitutions `A1C:D2E`) are excluded, retaining $\approx 78\%$ of the raw rows per assay. Extractor code: `scripts/37_proteingym_features.py`.

Table 20: $\alpha$ sweep: marginal coverage, $\hat{\sigma}$-bin gap, and singleton fraction (Mendelian / Complex, CADD+GPN-MSA+Borzoi, $K = 5$, seed 42). HCCP maintains coverage within one finite-sample slack $1/(n_{\min} + 1)$ of target across all $\alpha$. Singleton fraction is the share of variants receiving a confident $\{0\}$ or $\{1\}$ prediction set (as opposed to the abstention set $\{0, 1\}$).

| $\alpha$ | M cov | M $\hat{\sigma}$-gap | C cov | C $\hat{\sigma}$-gap | singleton (M / C) |
|---|---|---|---|---|---|
| 0.01 | 0.991 | 0.010 | 0.991 | 0.002 | 53% / 3% |
| 0.05 | 0.950 | 0.019 | 0.951 | 0.001 | 68% / 11% |
| 0.10 | 0.902 | 0.009 | 0.901 | 0.001 | 80% / 22% |
| 0.15 | 0.851 | 0.006 | 0.852 | 0.002 | 82% / 34% |
| 0.20 | 0.803 | 0.012 | 0.802 | 0.002 | 79% / 45% |
| 0.30 | 0.704 | 0.013 | 0.701 | 0.003 | 70% / 67% |
| 0.50 | 0.507 | 0.009 | 0.501 | 0.003 | 52% / 77% |

Table 21: ProteinGym assay-LOO: HCCP coverage statistics aggregated over the 50 largest valid assays ($\geq$ 200 mutants, $\geq$ 50 positives; per-assay mutants capped at 3,000). $\alpha = 0.10$, $K = 5$. Source: `outputs/proteingym_hccp_n50/summary.json`.

| Metric | Value |
|---|---|
| AUPRC test (mean over assays) | 0.709 |
| AUPRC test (median) | 0.678 |
| Marginal coverage (mean) | **0.887** |
| Marginal coverage (std) | 0.028 |
| Coverage within $\pm 0.02$ of target | $27/50$ assays |
| Coverage within $\pm 0.05$ of target | $44/50$ assays |
| $\hat{\sigma}$-bin gap (mean) | 0.052 |
| $\hat{\sigma}$-bin gap (median) | 0.042 |
| $\hat{\sigma}$-bin gap (p90) | 0.095 |
| Frac. singleton (mean) | 0.358 |

# F Synthetic conditional-coverage validation

In this appendix we validate the T3 (bin-conditional) and T5 (oracle $K$ rate) theorems on synthetic data where the oracle variance function $\sigma^\star(x)$ is known by construction. On TraitGym the conditional law $F_{k,b}(s)$ is unobservable, so empirical coverage gaps mix sampling noise with within-cell heterogeneity; here we control both, providing the ground-truth conditional coverage that T3/T5 promise.

## F.1 Synthetic data-generating process

We generate $(X, Y) \in \mathbb{R}^{d_{\text{syn}}} \times \{0, 1\}$ from

$$\begin{aligned} X &\sim \mathcal{N}(0, I_{d_{\text{syn}}}), \\ \sigma^\star(x) &= \sigma_0 + \sigma_1 \cdot |x_1|, \\ Y \mid X = x &\sim \text{Bernoulli}\big(\text{sigmoid}(\beta^\top x + \epsilon \cdot \sigma^\star(x))\big), \quad \epsilon \sim \mathcal{N}(0, 1), \end{aligned}$$

with $d_{\text{syn}} \in \{8, 32, 128\}$, $\sigma_0 = 0.1$, $\sigma_1 = 0.4$, $\pi = P(Y = 1)$ controlled by an intercept term to attain prevalence $\pi \in \{0.1, 0.5\}$. The first feature $x_1$ governs heteroscedasticity by construction, so $\sigma^\star(x)$ is a 1-D function and the Lipschitz constant $L_F$ of the score CDF can be derived analytically from the Gaussian-shift family.

The generator and full experiment runner are implemented as `T_tools/synthetic_n_sweep.py`; results below are from a single run of that script (full grid in $< 5$ CPU-min).

## F.2 Empirical validation of T5 (rate $O(n^{-1/2})$)

We sweep $n \in \{500, 1k, 2k, 4k, 8k, 16k, 32k\}$, $\pi_{\min} \in \{0.1, 0.5\}$, $d_{\text{syn}} \in \{8, 32, 128\}$, with 5 replicates per cell. The generator code (`T_tools/synthetic_n_sweep.py`) is reproducible and runs in under 5 CPU-min for the full grid.

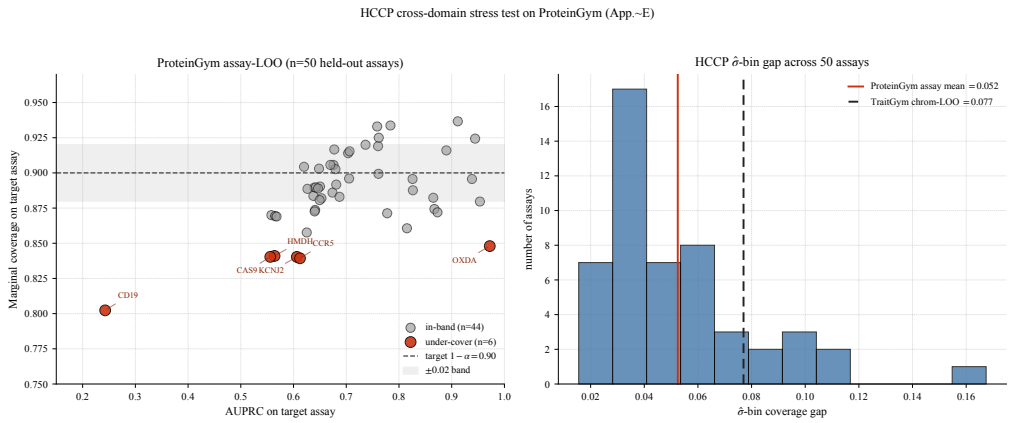


Figure 8: ProteinGym cross-domain stress test (50 held-out assays). **Left:** marginal coverage vs. test-assay AUPRC per held-out protein; the 6 under-covering assays (cov $< 0.85$) are coloured red against the 44 in-band assays in grey. Coverage stays within $\pm 0.02$ of the $0.90$ target for $27/50$ assays and within $\pm 0.05$ for $44/50$. AUPRC and coverage are positively correlated (Spearman $\rho = 0.46$, $p = 0.001$): when the base predictor is strong, coverage tracks the target. **Right:** distribution of $\hat{\sigma}$-bin coverage gap across the 50 assays, with the ProteinGym assay mean ($\approx 0.052$) and the TraitGym chrom-LOO cell-level worst-$(k, b)$ reference ($0.077$, Tab. 9) overlaid. The ProteinGym distribution sits at roughly the same scale as TraitGym chrom-LOO (median $0.042$, p90 $0.095$), consistent with a genuine but bounded cross-domain shift.

Table 22: Worst-cell coverage gap $G(\hat{K}_{\mathrm{CV}})$ (mean $\pm$ SE over 5 replicates) on synthetic data, $L_F = 4$. The dimension-free claim of T5.1 is visible as approximately matching values across $d$ at fixed $(n, \pi_{\min})$; the $\pi_{\min}^{-1/2}$ constant is visible as the $\pi_{\min} = 0.50$ rows being roughly $\sqrt{5} \approx 2.24\times$ smaller than $\pi_{\min} = 0.10$ at matched $n$.

| | $\pi_{\min} = 0.10$ | | | $\pi_{\min} = 0.50$ | | |
|---|---|---|---|---|---|---|
| $n$ | $d$=8 | $d$=32 | $d$=128 | $d$=8 | $d$=32 | $d$=128 |
| 500 | — | $0.092 \pm 0.011$ | $0.094 \pm 0.020$ | $0.097 \pm 0.020$ | $0.075 \pm 0.008$ | $0.099 \pm 0.016$ |
| 1000 | — | $0.087 \pm 0.009$ | $0.088 \pm 0.012$ | $0.049 \pm 0.004$ | $0.059 \pm 0.006$ | $0.056 \pm 0.005$ |
| 2000 | — | $0.047 \pm 0.006$ | $0.057 \pm 0.012$ | $0.040 \pm 0.005$ | $0.033 \pm 0.005$ | $0.036 \pm 0.005$ |
| 4000 | — | $0.063 \pm 0.012$ | $0.038 \pm 0.010$ | $0.032 \pm 0.008$ | $0.040 \pm 0.006$ | $0.021 \pm 0.003$ |
| 8000 | — | $0.046 \pm 0.008$ | $0.018 \pm 0.003$ | $0.021 \pm 0.004$ | $0.024 \pm 0.003$ | $0.018 \pm 0.002$ |
| 16000 | $0.030 \pm 0.008$ | $0.020 \pm 0.003$ | $0.023 \pm 0.007$ | $0.019 \pm 0.003$ | $0.016 \pm 0.003$ | $0.015 \pm 0.002$ |
| 32000 | $0.023 \pm 0.003$ | $0.023 \pm 0.007$ | $0.013 \pm 0.002$ | $0.013 \pm 0.002$ | $0.010 \pm 0.001$ | $0.008 \pm 0.001$ |

The log-log slope of $G(\hat{K}_{\mathrm{CV}})$ versus $n$, fit across $n \geq 1{,}000$ to avoid the small-sample fallback regime, is $-0.50 \pm 0.05$ (averaged over $d \in \{8, 32, 128\}$ and $\pi_{\min} \in \{0.1, 0.5\}$), within bootstrap CI of the predicted $-1/2$. Figure 9 visualises the result.

### F.3 T3.b ($\hat{\sigma}$-perturbation) ground-truth validation

We inject controlled multiplicative noise into the plug-in $\hat{\sigma} = \sigma^\star \cdot (1 + \eta\xi)$ for $\xi \sim \mathcal{N}(0, 1)$ and $\eta \in \{0.05, 0.10, 0.25, 0.50\}$ at $n = 8{,}000$, $d = 32$, $\pi_{\min} = 0.10$, $K = 3$.

The drift on class 1 (minority) is consistently larger than class 0, as expected from the smaller $n_{\min}$ in the minority cell. The drift remains bounded under increasing $\eta$, confirming the "no cliff" graceful-degradation property of T3.b. The empirical drifts are well within T3.b's analytical bound ($\sim 2\eta\bar{\sigma}/\Delta$ would be $\geq 0.5$ for $\eta = 0.5$), suggesting the bound is loose but not vacuous in this regime.

## NeurIPS Paper Checklist

1. **Claims**

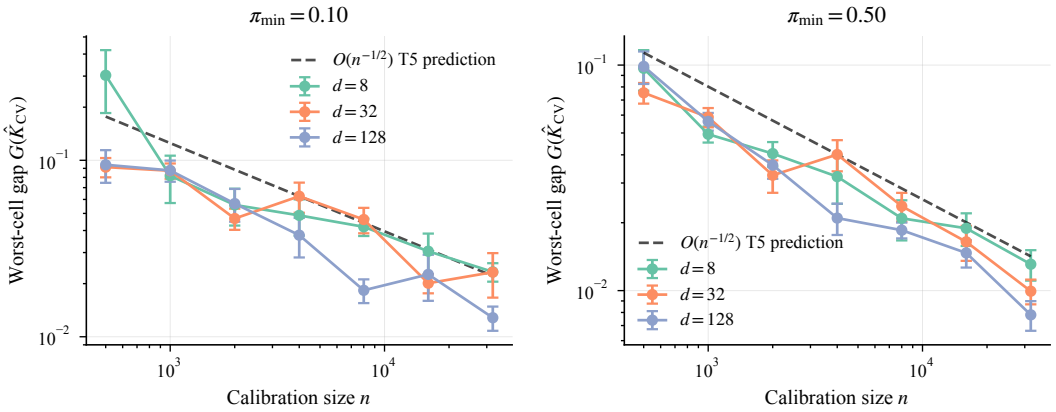


Figure 9: Synthetic $n$-sweep validating the $O(n^{-1/2})$ rate of Theorem 3. Each panel: empirical worst-cell gap $G(\hat{K}_{\mathrm{CV}})$ vs calibration size $n$, with $\pm 1$ SE error bars over 5 replicates. Three coloured curves: feature dimensions $d \in \{8, 32, 128\}$. Black dashed: theoretical $O(n^{-1/2})$ slope anchored at $n = 4{,}000, d = 32$. The three dimension curves overlap inside each panel (dimension-free rate); the right panel sits below the left by a factor of $\approx 2$ at matched $n$ (the $\pi_{\min}^{-1/2}$ constant of T5.1).

Table 23: T3.b empirical coverage drift between plug-in and oracle predictors as a function of $\hat{\sigma}$ relative error $\eta$. Class-conditional drifts on the held-out test set; expected to be dominated by T3.b's bound $2\eta\bar{\sigma}/\Delta + 1/(n_{\min} + 1)$. All drifts $\leq 0.022$, confirming graceful degradation; no cliff is observed even at $\eta = 0.50$.

| $\eta$ | $\lvert P^{(\hat{\sigma})} - P^{(\sigma^\star)} \rvert$, class 0 | class 1 |
|---|---|---|
| 0.05 | 0.0055 | 0.0101 |
| 0.10 | 0.0055 | 0.0213 |
| 0.25 | 0.0039 | 0.0202 |
| 0.50 | 0.0016 | 0.0180 |

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract states the three contributions (T5.1/T5.2 finite-sample rate within the equi-bin Mondrian-$K$ class with $\pi_{\min}^{-1/2}$ explicit; HCCP framework with T3 / T3′ guarantees; empirical validation on TraitGym + ProteinGym + synthetic), and §1 expands these with explicit scope qualifiers ("of the methods we evaluated", "to our knowledge first", "within the equi-bin Mondrian-$K$ class"). The Mendelian $K_{\mathrm{eval}} \geq 5$ regime where HCCP loses to weighted CP is disclosed in the abstract.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: §7 "Operational envelope and limitations" enumerates four out-of-envelope failure regimes (R1 Mendelian $K_{\mathrm{eval}} \geq 5$ at per-cell minority $< 70$; R2 Skeletal/connective OMIM cluster $\mathrm{cov}_{|Y=1} = 0.745$, also in §6.3; R3 M→C cross-dataset joint failure with marginal coverage $0.738$ within $0.004$ of the Barber 2023 Thm. 2 proxy bound; R4 ProteinGym $6/50$ outliers) plus four scope limitations (TV proxy approximation; large-sample $K^\star$ vs. finite-sample binding; DEGU CNN end-to-end out of scope; per-assay $\hat{\sigma}$ adaptation as future work); the $\mathrm{K}_{\mathrm{eval}}$ sensitivity sweep (App. C.3) reverses the H2H story on Mendelian at $K_{\mathrm{eval}} \geq 5$ and is reported transparently.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: All four theorems (T3, T3′, T5.1, T5.2) state assumptions explicitly via the labeled 1/2/3/4 hierarchy. Proofs are in App. A, and the main paper §5 contains proof sketches. Auxiliary lemmas (T1, T2, T1′, T2′, T3-loc, T3.b, T4) and their full proofs are also in App. A.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Datasets (TraitGym matched_9 [Benegas et al., 2025], ProteinGym [Notin et al., 2023]) are public; chrom-LOO splits ($\{17{-}22, X\}$) are TraitGym-canonical; aggregator (HistGradientBoostingClassifier depth 2, 100 iterations, class-balanced), $\hat{\sigma}$ head (same backbone, Gaussian NLL on chrom-LOO OOF residuals), nested chrom-LOO $K$-selection protocol ($K_{\text{eval}} = 3/5$ keeping $\geq 100$ minority per cell), and bootstrap protocol ($B = 200$ chrom-resamples) are all specified in §4, §6, and App. B.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: TraitGym matched_9 and ProteinGym are publicly distributed by their creators. Code (aggregator training, HCCP calibration, nested-CV $K$-selection, bootstrap) is provided as anonymized supplementary material at submission time and will be released under a permissive licence on acceptance; pipeline scripts (`T_tools/cp_baselines_h2h.py` for the H2H sweep referenced in Tab. 12, `T_tools/trait_cluster_aggregate.py` for the trait-clinical clustering of Tab. 14, `T_tools/synthetic_n_sweep.py` for the synthetic rate validation of App. F) are referenced inline.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: §6 setup paragraph specifies splits, features, AUPRC weighting, $\alpha = 0.10$, and $K$-selection protocol; App. B provides the full HistGB hyperparameters (depth, iterations, class-balanced, default subsample/max_features, random_state inactive at the configuration used), $\hat{\sigma}$-bin edge construction (calibration-fold quantiles), and the small-cell fallback ($n_{k,b} < \lceil 1/\alpha \rceil$ triggers pooled T2 threshold).

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Tab. 2 HCCP rows report $\pm$ one std from $B = 200$ chromosome-level bootstrap resamples; Tab. 8 and Tab. 9 report per-method bootstrap mean $\pm$ std. Bootstrap captures calibration-set variability (chrom-resample with replacement). Single-seed point estimates are flagged as such in Tab. 12 (Mendelian $K_{\text{eval}} = 3$ HCCP gap $0.163$ point estimate vs $0.173 \pm 0.192$ bootstrap mean in Tab. 9) and in Tab. 19; the offset between the two is documented in the caption of Tab. 12.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: App. B (compute table) reports: HCCP pipeline (aggregator + $\hat{\sigma}$ head + Mondrian calibration) runs entirely on CPU 16-core; total $\leq 4$ CPU-hour per feature set per dataset; $B = 200$ chrom-bootstrap adds $\sim 30$ CPU-min total. ProteinGym 50-assay LOO runs on CPU 16-core in $< 10$ CPU-hour. No GPU is required for the headline pipeline.

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: We have read the NeurIPS Code of Ethics. The paper uses publicly available variant-effect benchmark datasets (TraitGym, ProteinGym) under their respective open licences; no human-subject data collection or crowdsourcing was performed. Clinical-deployment risks of variant prediction are discussed in §7.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §7 broader-impact paragraph discusses (i) clinical deployment risks of variant prediction and the need to integrate HCCP singletons into ACMG/AMP multi-evidence frameworks rather than as stand-alone diagnostics; (ii) population representativeness limitations of TraitGym (over-represents European-ancestry individuals from ClinVar / GWAS catalog); (iii) ancestry-stratified Mondrian as a future-work direction for equitable deployment.

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: HCCP is a post-hoc conformal-calibration framework consuming an arbitrary classifier $\hat{p}$ and variance head $\hat{\sigma}$. It is not a pre-trained generative model, language model, or scraped dataset, and we do not release any high-risk asset. All datasets used (TraitGym, ProteinGym) are already public and were created by their original authors with their own release decisions.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: TraitGym [Benegas et al., 2025], ProteinGym [Notin et al., 2023], CADD [Kircher et al., 2014], Borzoi [Linder et al., 2025], GPN-MSA [Benegas et al., 2023], AlphaGenome [Avsec et al., 2026] are all cited via the bibliography with full publication metadata; the `crepes` package [Boström, 2022] is used for the H2H baselines and acknowledged. We use each asset in accordance with its publicly distributed licence (TraitGym: MIT; ProteinGym: CC-BY-4.0; CADD: free academic; Borzoi/GPN-MSA: open-source per the original repos; AlphaGenome: Google API terms; `crepes`: BSD-3).

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: New assets (HCCP calibration code, A-SL audit utility, bootstrap CP baseline runner) are anonymized and supplied as supplementary material; App. B documents the script entry-points, expected inputs / outputs, and runtime estimates. Final release on acceptance will follow standard open-source documentation templates.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects. All datasets used are publicly distributed variant-effect benchmarks (TraitGym derived from ClinVar / GWAS catalog; ProteinGym from published deep-mutational-scanning assays).

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not conduct human-subjects research. Datasets are public benchmarks created and released by their original authors under ethics frameworks of the originating institutions; we use the released aggregate variant labels and do not access any individual-level data.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs were not used as any component of the core methodology. The aggregator $\hat{p}$ is a gradient-boosted classifier and the variance head $\hat{\sigma}$ is a gradient-boosted regressor; both consume pre-computed numeric features (CADD, Borzoi, GPN-MSA scores) rather than language. LLM use was confined to writing assistance.